INFORMATION DYNAMICS AND BANK RUNS

An Event Study of the 2023 Banking Crisis

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Abstract

The 2023 banking crisis was triggered by the collapse of Silicon Valley Bank (SVB) on March 10. It showed how rapidly information spreads in the digital age, and how that information shapes depositor and investor behaviour. This thesis examines the information dynamics surrounding the crisis. The method is an event study applied to 48 publicly traded U.S. banks over April 2022 to June 2023.

Five crisis event dates are used. The market model is estimated over $[-210, -11]$ trading days and the event window is $[-2, +2]$. Cumulative Abnormal Returns (CARs) are computed for each bank. They are then related to balance-sheet fundamentals from FDIC Call Reports (Q4 2022): uninsured deposit ratios, asset size, Tier 1 leverage and loan-to-deposit ratios. Google Trends search volume proxies public attention. Returns are continuously compounded, so all CARs are cumulative log-return abnormal returns.

The five-day window around the SVB closure also contains the March 8 disclosure and the March 13 session on which the systemic-risk intervention was first priced. Over that window the naive Cumulative Average Abnormal Return is -22.31%, cumulated day by day across the sample. The mean of the 48 individual bank CARs is -21.78%. The two differ only because the number of banks contributing falls during the window, and Section 5.2 reconciles them. The figure measures the reaction to the March cluster rather than to the closure alone. Losses at the failed banks were extreme (SIVB -89.0%, FRC -104.5%, SBNY -33.8%). A log CAR of -104.5% corresponds to roughly -65% in simple-return terms. The banks all reacted to the same events on the same days, so their abnormal returns are not independent draws. The aggregate conclusion rests on a cross-sectional-dependence adjustment. Under that adjustment the sector-wide abnormal return is not statistically distinguishable from zero ($t = -0.77$). A complementary calendar-time portfolio regression does reject (-2.71% per day, $p = 0.002$), but it rests on fifteen crisis days. The two procedures disagree on significance, so the aggregate question is not treated as settled. What they share is that naive inference overstates the evidence. Only the portfolio specification supplies a second point estimate, and it is smaller than the naive figure. The dependence adjustment changes the test statistic and leaves the point estimate where it was.

The central result is cross-sectional. OLS finds the uninsured deposit ratio negatively associated with CARs. These banks are exposed to one common shock, so HC3 standard errors alone would not be an adequate basis for inference. The coefficient is therefore also tested by randomization inference. That procedure permutes the uninsured ratio across banks while holding the observed CAR vector and its dependence structure fixed. No permutation in either specification produced a coefficient as large in absolute value as the observed one. This is a sensitivity check rather than design-based randomization inference, since the ratios were not randomly assigned, and the HC3 figures remain the ones reported. The procedure addresses cross-sectional dependence only. It permutes the uninsured ratio alone, so it does not preserve that variable's correlation with size, and exchangeability says nothing about omitted variables. The estimates remain associations rather than causal effects throughout. The coefficient is -0.0112 in the parsimonious specification (95% CI $[-0.0185, -0.0038]$, HC3 $p = 0.004$). It is -0.0115 in the full model, which is the preferred specification and also has the higher adjusted R^2 (0.487 against 0.424), and it is itself significant at the 5% level ($p = 0.015$). A 10 percentage-point higher uninsured ratio corresponds to roughly 11 percentage points more negative CAR. Critically, the association survives excluding the three institutions that failed ($\beta = -0.0103$, $p = 0.023$). It is therefore not an artefact of the banks that collapsed. Quantile regression nonetheless shows the relationship is strongest in the lower tail. It declines from -0.0216 at the 10th percentile to -0.0016 at the 90th, and it disappears entirely once the worst-affected fifth of the distribution is removed. The contribution is to document that this cross-sectional association between uninsured funding exposure and crisis-period equity returns is robust but markedly heterogeneous, and concentrated among the more severely affected institutions. Coordination models near a fragility threshold would generate such a pattern. Direct tests for a threshold are inconclusive, so it is documented as an empirical regularity rather than an identified discontinuity. Bank size is insignificant and the G-SIB flight-to-safety premium is confounded with size. A two-way fixed-effects panel tests whether that association was larger on high-attention days. It returns the predicted negative sign but cannot distinguish it from zero (interaction -0.0044, wild cluster bootstrap $p = 0.31$). With fifteen date clusters the test has little power, and this is reported as a null.

These findings contribute to the literature on social media, information dynamics and bank fragility. They also carry implications for deposit insurance policy and banking supervision.

Keywords: bank run, event study, Silicon Valley Bank, abnormal returns, uninsured deposits, information dynamics, 2023 banking crisis

1 Introduction

1.1 Background and motivation

On March 10, 2023, Silicon Valley Bank (SVB) was closed by the California Department of Financial Protection and Innovation, and the Federal Deposit Insurance Corporation (FDIC) was designated as receiver. It was the second-largest bank failure in U.S. history at that time. Within 48 hours, on March 12, Signature Bank of New York was taken into receivership. On the first day of May, news emerged that First Republic Bank, one of the 15 largest banks by assets in the United States, had collapsed because of catastrophic deposit outflows. First Republic was the larger of the two, so it took second place in that ranking and pushed SVB down to third. Regulators, academics and market participants alike were stunned by the remarkable speed of the 2023 banking crisis.

Information played a role in this crisis, and that sets it apart from earlier episodes. The collapse of SVB was characterised at the time as a crisis of information transmission. Once SVB disclosed heavy unrealised bond-portfolio losses on the evening of March 8, the market witnessed a rapid succession of events: surging social-media commentary, rapid outflows of uninsured deposits and a severe downward repricing across regional banks. Whether information transmission drove this sequence or simply accompanied it, this thesis cannot establish, and no such claim is made. While this narrative provides the motivating background, Section 3.7 clarifies the scope and the limitations of the research design. Depositors did not have to be physically present at a branch to pull their deposits. A mobile banking app made withdrawal much easier with a few taps, and depositors moved their funds while informed in real time by tweets, financial newsletters and venture-capital WhatsApp groups. In the language of Diamond and Dybvig (1983), the event has the form of a self-fulfilling panic, and what made it special was how fast news about the weaknesses of the banks spread through digital channels rather than through physical queues at branches. Depositor-level data would be required to prove that digital communication directly caused this coordination. However, this thesis does not have access to that information.

The information dynamics of the 2023 banking crisis are studied here with event study methodology, and three things are asked. First, how bank equities moved around the five major crisis events, and how far that movement was concentrated in a few institutions. Second, whether the cross-section of returns can be explained by bank-level vulnerability to information-driven runs, and in particular by uninsured deposit ratios. Third, whether public information acquisition, measured by Google Trends search volume, can be used to describe the information channel. One distinction is kept throughout and is worth stating here. The average bank in the sample recorded a large negative abnormal return. However, under the dependence-adjusted test that Section 5.2 treats as primary, that sector-wide average cannot be shown to differ from zero. A large observed average loss and a statistically established sector-wide abnormal return are two different claims, and only the first one is made here.

1.2 Research questions

This thesis asks three research questions.

RQ1: How did U.S. bank equities respond to the five key events of the 2023 banking crisis, and how far was that response concentrated in a few institutions rather than shared by the sector? The form of the question is deliberate. Under the dependence-adjusted test that Section 5.2 treats as primary, the sector-wide average is not distinguishable from zero, so a simple yes-or-no test of aggregate significance would misdescribe the data. What is informative is how the reaction was distributed across banks.

RQ2: Does a bank's uninsured deposit ratio explain cross-sectional variation in CARs, once asset size, Tier 1 leverage and the loan-to-deposit ratio are controlled for?

RQ3: How does public search activity, measured by Google Trends, move in relation to the timing of the observed abnormal returns? This question is posed as a descriptive one. The analysis documents stylised facts about attention rather than testing a causal mechanism. A worldwide daily search index cannot be assigned to individual banks, and it cannot be separated from the arrival of crisis news, so it can show co-movement between attention and abnormal returns but cannot identify an information channel. The panel of Section 5.4 sharpens the description by using cross-sectional variation in funding fragility. It does not remove these limitations, and no causal claim is made from it.

1.3 Contribution and structure

The question behind this thesis is whether information-sensitive funding structures, and in particular high concentrations of uninsured deposits, amplified market reactions to the 2023 crisis beyond what ordinary balance-sheet fundamentals would predict. It is not simply whether weaker banks fell further. It is where in the cross-section the market attached a penalty to run-prone funding, and how concentrated that penalty was. The answer, established in Chapter 5, is that the penalty was real but almost entirely confined to the lower tail of the return distribution. Coordination models of bank runs would produce such a pattern near a fragility threshold, although the design cannot establish that a threshold is what produced it.

Three contributions follow. First, the classical event-study framework is applied to 48 publicly traded U.S. banks from the KBW regional and national indices. The sample covers the three failed institutions (SIVB, SBNY, FRC), the severely repriced survivors (PACW, WAL) and other stressed regionals (ZION, KEY, RF, CMA), and the five largest banks in the sample (JPM, BAC, WFC, USB, TFC), of which the first three are the G-SIBs. The groups are defined in Table 2 and used in the same way thereafter. This cross-section is substantially broader than that of earlier single-event studies of the crisis, and it is analysed with inference procedures suited to perfectly clustered events. Second, the cross-sectional return responses are linked directly to balance-sheet fundamentals from the FDIC, and the resulting relationship is described across the whole conditional distribution rather than at the mean alone. Third, Google Trends attention data are related to the event-study results, which gives a descriptive account of the information dynamics that coordination models (Goldstein and Pauzner 2005) emphasise.

Two limits should be stated at the outset. All banks are observed in a single crisis, with no untreated control group and no exogenous variation in deposit-insurance coverage, so what the analysis identifies is how the equity market priced the run-risk carried by funding structures. That is a market-pricing statement rather than a causal effect. And the search data are worldwide and aggregate, so they support a description of the information channel rather than a test of it.

The rest of the thesis is organised as follows. Chapter 2 reviews the theoretical and empirical literature. Chapter 3 describes the methodology. Chapter 4 presents the data. Chapter 5 reports the empirical results. Chapter 6 concludes.

2 Literature review

2.1 Bank run theory: panic versus fundamentals

The seminal model of Diamond and Dybvig (1983) establishes that bank runs can be self-fulfilling equilibria. Banks finance illiquid long-term assets with liquid demand deposits. Any signal, however noisy, that other depositors are withdrawing can therefore rationally induce withdrawal. The model predicts two equilibria. In the good equilibrium depositors keep funds on deposit. In the bank-run equilibrium all depositors withdraw regardless of fundamentals. Government deposit insurance can eliminate the bad equilibrium by removing the first-mover advantage of early withdrawal.

Goldstein and Pauzner (2005) remove this indeterminacy by changing the informational environment rather than the contract. Fundamentals are stochastic, and depositors observe them not in common but through slightly noisy private signals. This global-games perturbation eliminates the multiplicity. It delivers a unique Bayesian equilibrium in which a run occurs if and only if fundamentals fall below a critical threshold θ^*. The result is often described as an intermediate position between panic and insolvency, but the model's claim is sharper than that. Runs remain panic-based, in the sense that in most run states a depositor withdraws only because she expects others to. What fundamentals do is not to determine actions directly but to "serve as a device that coordinates agents' beliefs on a particular outcome" (Goldstein and Pauzner 2005, 1295). Fundamentals select the equilibrium, and expectations execute it.

Two features of that equilibrium shape the empirical design of this thesis. First, the state space partitions into three regions rather than two (Goldstein and Pauzner 2005, 1304): a lower region where withdrawal is dominant and runs are fundamental-based, an intermediate region where runs occur but are panic-based, and an upper region where no run occurs. Fundamentals discriminate between outcomes only near the boundary of the intermediate region, so the association between measured fragility and market outcomes should be steep among institutions close to that boundary and flat among those comfortably above it. Section 5.5 examines exactly this across the conditional return distribution. Second, the threshold is not a primitive but a function of the banking contract. $\theta^*(r_1)$ is increasing in the short-term payment the bank promises, so an institution offering more liquidity on demand is vulnerable over a wider range of fundamentals (Goldstein and Pauzner 2005, 1309). Fragility is therefore partly a property of funding structure, and to that extent observable, and Section 3.4 takes the uninsured deposit ratio as its empirical counterpart. The model's welfare results are not tested here.

Calomiris and Kahn (1991) supply the microfoundation for why uninsured depositors should be the marginal actors in such a run. Demandable debt serves an informational role, since depositors who withdraw first are paid in full and therefore have an incentive to monitor bank quality. The corollary is central to the design here: the depositors with the strongest monitoring incentive are precisely those with most to lose from delay, namely large uninsured balances. The uninsured deposit ratio is therefore not merely a measure of how much funding could flee, but of how much is held by agents with both the incentive to acquire information and the incentive to act on it first. That double role is why it is the primary cross-sectional regressor in Section 3.4.

Against this coordination-failure tradition stands a substantial and recently reinvigorated challenge. Correia, Luck, and Verner (2026) assemble the longest micro-level panel of U.S. commercial bank balance sheets to date (1863–2024) and document that failing banks are characterised by rising asset losses, deteriorating solvency and growing reliance on expensive non-core funding, so systematically that failure is highly predictable from simple accounting metrics. Even historical failures accompanied by runs are strongly related to weak fundamentals, and low recovery rates on failed banks' assets indicate that most failed institutions were fundamentally insolvent. They therefore cast doubt on the empirical importance of purely panic-driven runs.

This debate is not incidental to the present study. The empirical results speak directly to it. If the coordination view is right, a common information shock should reprice the sector broadly. If the fundamentals view is right, the repricing should be concentrated in the institutions whose balance sheets were already compromised. As Chapter 5 shows, the 2023 evidence contains elements of both. Repricing is concentrated in the banks with the weakest funding structures, as the fundamentals view predicts. But its tail-concentrated form is what coordination models emphasise, even though the design cannot confirm a threshold mechanism. Section 6.1 returns to this.

2.2 Information asymmetry and banking fragility

Chari and Jagannathan (1988) model bank runs driven by asymmetric information. Some depositors observe a bad signal about bank quality and withdraw. Uninformed depositors observe the long queue, infer that something is wrong, and also withdraw. This mechanism generates contagion. A run at one bank can trigger runs at other banks, even fundamentally sound ones, if depositors cannot distinguish between individual and systemic information. It supplies the theoretical reason to expect abnormal returns at banks with no direct exposure to SVB, and it is also the mechanism against which the steep cross-sectional gradient documented in Section 5.1 is best assessed. Indiscriminate contagion predicts uniform markdowns, whereas informative contagion predicts sorting by observable fragility.

Gorton (1988) provides historical evidence that bank panics in the pre-Federal Reserve era were triggered by news about aggregate economic conditions rather than idiosyncratic bank failures. That aggregate-information channel yields a testable prediction here: systemically framed events, such as the systemic risk exception of March 12, 2023, should generate widespread abnormal returns across all banks. Sections 5.1 and 5.2 find it only partially borne out, since once cross-sectional dependence is corrected the sector-wide component is significant only under the less conservative of the two corrections, and the response is dominated by a few severely affected institutions. The Gorton benchmark is useful precisely because the 2023 episode departs from it.

More recently, Jiang, Matvos, Piskorski, and Seru (2024) approach the same fragility from the asset side. They mark U.S. bank assets to market for the rate increases of 2022 and early 2023 and find an average decline of about 10 per cent, some $2 trillion in aggregate. In an accompanying model, losses of this size can precipitate self-fulfilling solvency runs even when assets are perfectly liquid, and the variable determining exposure is uninsured leverage, since only uninsured claimants stand to lose in default. Their argument pulls in two directions for the present design. It supplies a structural reason to expect the uninsured share to predict crisis-period losses, and it also shows that the same variable indexes exposure to unrealised asset losses, so a negative coefficient cannot by itself distinguish coordination failure from fundamental insolvency. That ambiguity is the interpretive limit of the cross-sectional design, returned to in Sections 3.7 and 6.1.

Depositor-level evidence corroborates the mechanism. Iyer and Puri (2012), using account-level data from a bank that experienced a run, show that depositors with weaker relationships and larger uninsured balances are the first to flee and that social networks propagate withdrawals. Martin, Puri, and Ufier (2026) likewise find outflows concentrated among uninsured depositors following adverse regulatory news, but their central result complicates the picture: as the bank approached failure it replaced those outflows almost entirely with new insured deposits attracted by higher term rates. Gross uninsured outflows therefore overstate net funding pressure, and insured inflows weaken the discipline uninsured funding is supposed to impose. Egan, Hortaçsu, and Matvos (2017) embed this asymmetry in a structural model of deposit competition, showing that reliance on runnable uninsured deposits materially raises the probability of a solvency-driven run. Together these studies establish the uninsured share as the privileged measure of funding fragility, which motivates its use here and connects the study to the policy debate over whether coverage limits should be raised, made risk-based, or supplemented by liquidity requirements.

2.3 Social media, digital banking, and modern bank runs

The 2023 crisis occurred against a backdrop of unprecedented connectivity. Cookson et al. (2026) provide the most direct evidence to date that social media activity preceded and amplified the bank run. Using Twitter data, they show that pre-existing Twitter attention and exposure, rather than negative sentiment per se, predicted larger stock market losses and deposit outflows during the run. High-exposure banks lost roughly 4.3 percentage points more market value over the run window. The fact that attention, rather than tone, is the operative variable is theoretically significant. It is consistent with the Goldstein and Pauzner (2005) mechanism, in which what matters is the commonality and precision of the public signal, its capacity to synchronise beliefs about what others will do, rather than whether the signal is favourable or unfavourable.

Bales and Burghof (2024) examine the same channel at higher frequency, using intraday Twitter and Google search attention during SVB's collapse. They find bidirectional lead-lag relationships between public attention and SVB's returns, with attention spikes preceding significant price declines. They attribute a role to both attention and Twitter sentiment in accelerating the crash. Their study is the closest methodological precedent for the search-attention proxy employed in Section 3.6. Note that the two studies diverge on the role of tone. Cookson et al. isolate attention as the operative channel and find sentiment does not amplify run risk, whereas Bales and Burghof assign sentiment an independent role. The present study cannot adjudicate this disagreement. Its worldwide search index measures attention volume, not valence, and the analysis here is accordingly framed in terms of attention alone.

Attention is only one half of the digital-era mechanism. The other is the infrastructure that converts attention into withdrawal within minutes. Benmelech, Yang, and Zator (2023) document that bank branch density declined sharply over the past decade, with U.S. branches falling from roughly 99,550 in 2009 to 79,186 in 2022 while deposits almost doubled between 2016 and 2022. Low-density, digitally intermediated banks grew fastest by offering superior digital service and higher rates, attracting large uninsured balances, and when conditions deteriorated in 2023 those same banks suffered heavy uninsured outflows and steep price declines. Their finding corroborates the uninsured-funding channel through an entirely independent variable, and it identifies branch density as the most promising complementary regressor for future work of this design.

Taken together, this literature motivates the approach adopted here, which constructs an information index from Google Trends data and relates it to abnormal stock returns.

2.4 Event study methodology in banking

MacKinlay (1997) provides the canonical reference for event study methodology in finance. The approach involves estimating a market model over an estimation window preceding the event, using the estimated parameters to construct expected returns over an event window, and computing abnormal returns as the difference between actual and expected returns. Under the null hypothesis of no abnormal information, abnormal returns should average zero.

The methodology's reliability with daily data was established by Brown and Warner (1985), whose simulation evidence remains the standard justification for the design used here. They show that with daily returns the simple market model performs well and is robust to non-normality of individual returns, so that more elaborate benchmark models yield little improvement in power. Their results carry two warnings that bear directly on this study. First, the properties of the test statistics depend heavily on whether abnormal returns are cross-sectionally independent. When event dates are clustered in calendar time, standard tests are badly misspecified and reject the null far too often. Second, event-period variance frequently exceeds estimation-period variance, so procedures assuming a constant residual variance overstate significance. Both conditions obtain in the present setting, where every sample bank shares the same event dates and volatility rises sharply during the event window.

Both warnings shape the design of Chapter 3. The Kolari and Pynnönen (2010) correlation adjustment answers the first and the Patell (1976) standardisation the second, and Section 3.3 sets out the Patell test and Section 5.2 the correlation adjustment. A difficulty specific to 2023 is that the key event dates fall in rapid succession (March 8, 10, 12, 15), so estimation and event windows risk overlapping. The conservative estimation window ending 11 trading days before each event ($[-210,-11]$) addresses this.

2.5 The 2023 banking crisis

The rapid rise in interest rates through 2022 created large unrealised losses on bank bond portfolios. These became publicly visible when SVB disclosed them on March 8. The FDIC reported that aggregate unrealised losses on securities across U.S. banks exceeded $620 billion at end-2022. Jiang, Matvos, Piskorski, and Seru (2024) show that marking banks' asset portfolios to market revealed sharp declines that left institutions with high uninsured-deposit shares, SVB among the most exposed, especially fragile.

Drechsler et al. (2026) focus on the deposit channel. They show that banks with high uninsured deposit ratios experienced disproportionately large deposit outflows in early 2023, and that these outflows were concentrated among banks that had not hedged their interest-rate exposure. Their theoretical contribution is equally relevant. Banks pay below-market rates on deposits, so the deposit franchise is itself a valuable and runnable asset whose value rises with the level of interest rates. Runs therefore become both more damaging and more likely precisely when rates are high. Their findings are consistent with the hypothesis tested in this thesis, that uninsured deposit ratios predict cross-sectional variation in market-assessed vulnerability.

The systemic risk exception invoked on March 12, announced jointly by the Treasury, the Federal Reserve and the FDIC, covered all depositors at SVB and Signature Bank regardless of the $250,000 limit. Ex ante it might therefore have been expected to generate positive abnormal returns, as markets re-priced the probability of systemic support. Whether it did so is an empirical question, examined in Section 5.1.

3 Methodology

3.1 Event study design

This study uses the standard event study methodology of Brown and Warner (1985) and MacKinlay (1997) to measure how the equity of U.S. banks reacted to five distinct events during the 2023 crisis. Event studies are the usual tool in empirical corporate finance for measuring how the market revalues a firm once specific news arrives. The idea behind them is that in an informationally efficient market the full price reaction to new public information comes quickly, and can therefore be captured in a short window around the announcement.

Five event dates are examined. Each one is a distinct stage in the information cascade, and Table 1 gives them with their trading dates. March 12, 2023 fell on a Sunday, so day 0 for Event 3 is the nearest trading day, Monday March 13, which was the first session on which the announcement could be priced. Keeping the events separate makes it possible to compare the market response to different kinds of information: fundamental news (Event 1), regulatory intervention (Events 2 and 3), and contagion and resolution signals (Events 4 and 5).

Event	Announcement	Day 0 (trading)	Description
1	Wed, March 8, 2023	March 8, 2023	SVB discloses bond-portfolio losses and announces capital raise
2	Fri, March 10, 2023	March 10, 2023	SVB closed by California regulator, FDIC appointed receiver
3	Sun, March 12, 2023	March 13, 2023	Signature Bank closed, systemic risk exception invoked
4	Wed, March 15, 2023	March 15, 2023	Contagion escalation: First Republic downgraded to junk, Credit Suisse crisis
5	Mon, May 1, 2023	May 1, 2023	First Republic seized by FDIC and sold to JPMorgan

Table 1: The five crisis event dates. For Event 3 the announcement fell on a Sunday, so day 0 is the next trading session.

The estimation window runs over trading days $[-210,-11]$ before each event, which gives about 200 observations for estimating the normal return parameters of each bank. The gap between the end of the estimation window (day -11) and the start of the event window (day -2) works as a buffer, so that market anticipation in the days just before the crisis does not contaminate the normal return estimates. A 200-day window is standard in the literature. It leaves enough degrees of freedom for stable OLS estimates while staying inside a single economic regime. The event window runs over trading days $[-2,+2]$, five days centred on the event date. The pre-event days (-2, -1) pick up information leakage or rumour, the event day (0) picks up the main price discovery, and the post-event days ($+1$, $+2$) pick up drift or delayed adjustment. A narrow ± 2 window is preferred to wider ones because the events came in rapid succession. March 8, 10, 12 and 15 all fall within one calendar week, so a longer window would be contaminated by the neighbouring events.

3.2 Market model and abnormal returns

The market model was introduced by Sharpe (1963) and put into an event study setting by Brown and Warner (1985). It relates the daily return of each bank to the market return through a simple linear regression estimated over the pre-event window. For bank i on trading day t in the estimation window, the model is:

$$R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it} \tag{1}$$

Here R_{it} is the continuously compounded (log) return of bank i on day t, computed as $\ln(P_{it}/P_{i,t-1})$ using adjusted closing prices to account for dividends and stock splits. R_{mt} is the daily log return of the S&P 500 index, which serves as the market portfolio proxy. α_i is a bank-specific intercept capturing average excess return relative to the market. β_i is the market beta, measuring the sensitivity of bank i's return to systematic market movements. ε_{it} is the idiosyncratic error term. The parameters are estimated by OLS over the estimation window.

The identifying assumption of the market model is that the error term satisfies $E[\varepsilon_{it}] = 0$ and $\mathrm{Cov}(\varepsilon_{it}, R_{mt}) = 0$. In other words, the market factor picks up all the systematic variation in bank returns, and whatever is left during the event window is bank-specific information. The model also assumes that the return-generating process is linear, that α_i and β_i are stable between the estimation and event windows, and that there is no thin trading. Stability is the most serious of these concerns here, because during a systemic banking crisis the betas of banks can move a good deal as the risk profile of the sector changes. This is a real limitation. It is addressed in part by the Fama-French robustness check of Section 3.5, which adds exposure to size and value factor returns, although on constant loadings.

The choice of the S&P 500 as benchmark needs a word, because 2023 was a shock to one sector rather than to the whole market. A broad index barely moved while bank equities fell sharply, so most of that fall is counted as abnormal. That is the intended reading, since what is of interest is the repricing specific to banks, and a bank index as benchmark would net out exactly the component being measured. The cost is that a bank's differential sensitivity to a banking-sector factor is absorbed into its abnormal return, and that sensitivity could itself be correlated with uninsured funding. Two checks bound this. Section 5.5 re-estimates against a value-weighted bank index and leaves the sign and significance pattern unchanged. The Fama-French CARs correlate at 0.998 with the market-model CARs. Neither is a full banking-sector factor model, which would be the natural refinement.

The Abnormal Return (AR) for bank i on event-window day τ is defined as the realised return minus the counterfactual normal return predicted by the estimated market model:

$$\mathrm{AR}_{i\tau} = R_{i\tau} - \left(\hat{\alpha}_i + \hat{\beta}_i R_{m\tau}\right) \tag{2}$$

A negative AR means the stock of the bank fell by more than the market model predicts for that day, given how the S&P 500 moved, and that is evidence of bad news specific to the bank. The Cumulative Abnormal Return (CAR) for bank i over the event window $[\tau_1, \tau_2] = [-2, +2]$ is:

$$\mathrm{CAR}_i(\tau_1, \tau_2) = \sum_{\tau=-2}^{+2} \mathrm{AR}_{i\tau} \tag{3}$$

For cross-sectional aggregation, the Average Abnormal Return (AAR) on day τ and the Cumulative Average Abnormal Return (CAAR) over the full event window, across the N_τ banks trading on day τ, are:

$$\mathrm{AAR}_\tau = \frac{1}{N_\tau}\sum_i \mathrm{AR}_{i\tau} \qquad \text{and} \qquad \mathrm{CAAR} = \sum_\tau \mathrm{AAR}_\tau \tag{4}$$

The variance of the CAR for bank i, accounting for estimation error in the market model parameters, is:

$$\sigma^2(\mathrm{CAR}_i) = (L+1)\,\hat{\sigma}_i^2 \left[1 + \frac{1}{M} + \frac{\left(\bar{R}_{m,\mathrm{event}} - \bar{R}_{m,\mathrm{est}}\right)^2}{\sum_t \left(R_{mt} - \bar{R}_{m,\mathrm{est}}\right)^2}\right] \tag{5}$$

Here $L = \tau_2 - \tau_1 = 4$, so that $L+1 = 5$ is the number of trading days in the $[-2, +2]$ window. M is the number of valid estimation-window observations, which is 200 for every bank in the sample: returns are computed on each series' own observed prices, so a day missing for one bank does not remove an observation from any other. $\hat{\sigma}_i^2$ is the estimated residual variance from the market model. The third term corrects for out-of-sample prediction error when the event-window market return deviates from its estimation-window mean. The implementation in Appendix A applies the first two terms only, $1+1/M = 1.005$, and omits the third, which at Event 2 equals 0.00024 and would move the standard error by about a hundredth of one per cent. For the halted banks the multiplier is the actual number of traded days rather than five. Written out, the quantity the code computes is

$$\hat{\sigma}(\mathrm{CAR}_i) = \hat{\sigma}_i \sqrt{n_i\left(1+\frac{1}{M}\right)} \tag{6}$$

where n_i is the number of event-window days on which bank i traded, which is five for every bank that traded throughout. This is the expression implemented in Appendix A.3. Finally, abnormal returns are defined as daily log returns, so cumulative abnormal returns are additive across days. A CAR may therefore fall below -100% without violating limited liability: a log CAR of x corresponds to a simple cumulative return of $e^x - 1$. First Republic's Event 2 CAR of -104.50% (log) thus maps to $e^{-1.0450} - 1 \approx -64.8\%$, a severe but finite equity loss.

3.3 Statistical tests

Testing whether the observed CARs can be distinguished from zero requires care in the choice of test statistic. Three approaches are used here, and each rests on different assumptions.

The simplest is the cross-sectional t-test of $H_0 : \mathrm{CAAR} = 0$, with

$$t_{cs} = \frac{\mathrm{CAAR}}{s(\mathrm{CAR}_i)/\sqrt{N}},$$

where $s(\mathrm{CAR}_i)$ is the cross-sectional standard deviation of the individual CARs. It handles heteroskedasticity across banks well enough, but it assumes that the banks are independent of one another, and that assumption fails when all N of them meet the same event on the same calendar date.

The Patell (1976) standardised abnormal return test addresses heteroskedasticity across banks. It scales each bank's CAR by its individual prediction error standard deviation before aggregating. The Standardised Cumulative Abnormal Return (SCAR) for bank i is:

$$\mathrm{SCAR}_i = \mathrm{CAR}_i/\hat{\sigma}(\mathrm{CAR}_i) \tag{7}$$

where $\hat{\sigma}(\text{CAR}_i)$ is the prediction-error-adjusted standard deviation from the formula in Section 3.2. Under the null of no abnormal returns, SCAR_i follows a t-distribution with $M-2$ degrees of freedom. The aggregate Patell Z-statistic is:

$$Z_p = \frac{\sum_i \text{SCAR}_i}{\sqrt{\sum_i v_i}}, \qquad v_i = \frac{M_i - 2}{M_i - 4} \tag{8}$$

The scaling factor v_i corrects for the heavier tails of the t-distribution in finite samples, and Z_p converges to $N(0,1)$ under the null as $N \to \infty$. The Patell test has more power than the simple t-test when the residual variances of banks differ a good deal, which is likely in a sample that runs from G-SIBs (low $\hat{\sigma}$) to small community banks (high $\hat{\sigma}$).

Both the cross-sectional t-test and the Patell Z assume that the abnormal returns of individual banks are independent of one another, and that assumption fails when every bank shares the same event date. MacKinlay (1997) notes that under calendar-time clustering the covariance terms in the variance of the CAAR are not zero, so ordinary standard errors are too small and the t-statistics too large. The clustering here is as extreme as it gets. All five events hit every bank in the sample at once, and the average pairwise correlation of abnormal returns is 0.616. The variance of the average abnormal return is then understated by exactly the sum of the omitted covariance terms, which is why the naive Event 2 statistics come out so large. For that reason the Kolari and Pynnönen (2010) correlation adjustment and the calendar-time portfolio are treated here not as optional robustness checks but as the inference procedures the design requires, and the naive statistics are kept only for illustration.

To address clustering, this study implements the calendar-time portfolio approach of Jaffe (1974) and Mandelker (1974), later formalised by MacKinlay (1997). On each calendar day t, a portfolio is formed consisting of all banks whose event window includes day t. The portfolio return P_t is regressed on the market return:

$$P_t = \alpha_p + \beta_p R_{mt} + \eta_t \tag{9}$$

The intercept α_p is the average daily abnormal return across the banks in the portfolio on that day. Because this is a single time-series regression, it deals with the correlation across banks on its own: all the banks sit in the same portfolio, so their co-movement nets out in the portfolio return. The t-statistic on α_p therefore gives inference that survives cross-sectional dependence, and the standard errors carry a Newey-West correction for serial correlation in the portfolio series. The number of time-series observations available is small, however, because it is limited to the distinct calendar days that fall inside at least one event window. The test is robust to dependence but weak in power, so on its own neither a rejection nor a failure to reject carries much weight. Section 5.2 reports the estimate on those terms, as a complement to the dependence adjustment rather than as the basis for the aggregate conclusion. The estimated daily portfolio alpha multiplied by 5, the length of the event window, gives the CAAR under this approach.

3.4 Cross-sectional regression

Event 2 (March 10, 2023) is the economically most important of the five. To find out which bank characteristics explain the cross-section of market reactions at that event, an ordinary least squares (OLS) regression of CARs on balance-sheet characteristics is estimated. The baseline specification is:

$$\mathrm{CAR}_i = \gamma_0 + \gamma_1 \mathrm{UninsuredDep}_i + \gamma_2 \log(\mathrm{Assets}_i) + \gamma_3 \mathrm{Tier1}_i + \gamma_4 \mathrm{LTD}_i + u_i \tag{10}$$

The variable of interest was chosen on theoretical rather than statistical grounds. In Goldstein and Pauzner (2005) the run threshold $\theta^*(r_1)$ rises with the short-term payment the bank promises on demand, so the liquidity written into the contract is itself the source of fragility (Section 2.1). The uninsured deposit ratio is the closest thing to that in Call Report data, because it measures the share of a bank's funding held by claimants who can withdraw at par and who, having no insurance, have every reason to do so first. Here the regressors are measured at Q4 2022 from FDIC Call Reports. $\mathrm{UninsuredDep}_i$ is the ratio of uninsured deposits to total deposits. $\log(\mathrm{Assets}_i)$ is the natural logarithm of total assets in billions. $\mathrm{Tier1}_i$ is the Tier 1 leverage ratio. LTD_i is the loan-to-deposit ratio. Additional specifications include a G-SIB dummy, equal to one for the three global systemically important banks in the sample (JPM, BAC, WFC), and a regulatory exemption dummy, equal to one for banks holding less than $100 billion in total assets. The Economic Growth, Regulatory Relief and Consumer Protection Act of 2018 released institutions below that threshold from enhanced prudential standards, including the Liquidity Coverage Ratio and supervisory stress testing. The dummy therefore identifies the group facing the lightest liquidity regulation going into the crisis, and 35 of the 48 sample banks fall below it.

OLS is the right estimator under the Gauss-Markov conditions, and four of them are worth a comment here. The 48 banks are all the sample institutions with usable price histories, the two KBW indices combined as described in Section 4.1, rather than the constituents of any single index. They are not a random draw, so the inference applies to this population and not to a wider one. Multicollinearity is moderate. On the widest specification estimated, log(Assets) carries the highest variance inflation factor at 6.22 and the G-SIB dummy 2.65. Neither is above the conventional threshold of 10, and the uninsured-deposit ratio, the variable of interest, is essentially unaffected (VIF = 1.38). Taking assets in levels rather than logs would raise the size variable to 13.2 and the G-SIB dummy to 11.4, both above the conventional threshold, which is one reason the logarithmic transformation is used. The G-SIB indicator nevertheless rests on three observations. Its coefficient is therefore imprecisely estimated for that reason rather than because of collinearity, and Section 5.3 reads it accordingly. Homoskedasticity is rejected (Breusch-Pagan LM = 21.16 on six degrees of freedom, $p = 0.002$). That does not bias the coefficients, but it does make ordinary standard errors invalid, which is why the robust estimator below is used. The identifying assumption, that the regressors are uncorrelated with the error term, cannot be tested. It is the substantive limitation discussed in Section 3.7, because omitted characteristics such as loan-portfolio quality or the strength of the deposit franchise could easily be correlated with both funding structure and crisis-period returns.

The theory of Section 2.1 says that fundamentals separate outcomes only near the run boundary, so the conditional mean is not the only thing worth looking at. The cross-section is therefore also estimated by quantile regression (Koenker and Bassett 1978), which fits the conditional τ-quantile by minimising a check-function loss instead of squared error and so gives a separate slope at each point of the distribution. Quantiles $\tau = 0.10$ through 0.90 are estimated by the Barrodale–Roberts simplex algorithm, which solves the quantile-regression linear programme exactly. Standard errors come from 400 bank-level bootstrap resamples, the analytic variance being unreliable at this sample size. Results appear in Table 7.

Under heteroskedasticity OLS stays unbiased, but its ordinary variance estimator is not valid. The HC3 heteroskedasticity-consistent (sandwich) estimator of MacKinnon and White (1985) is used instead. It corrects the White (1980) estimator for small-sample bias by inflating each squared residual by $(1-h_{ii})^{-2}$, where h_{ii} is the leverage of observation i:

$$\hat{V}_{\text{HC3}} = (X'X)^{-1}\left[\sum_i \left(\frac{\hat{u}_i}{1-h_{ii}}\right)^2 x_i x_i'\right](X'X)^{-1} \tag{11}$$

One limitation has to be stated here, because it is the same problem Section 3.3 raises for the aggregate tests. HC3 handles heteroskedasticity but assumes the observations are independent, and these banks are not. Their CARs are measured over one common window, with an average pairwise correlation of abnormal returns of 0.616 (Section 5.2). A shock that moves every CAR by the same amount is absorbed by the intercept and leaves the slope alone. But a shock that loads differently across banks in a way correlated with uninsured funding would leave the HC3 errors too small. It would be inconsistent to reject the naive aggregate statistics because of dependence and then rest the central result on an estimator with the same weakness. The coefficient is therefore also put through a permutation sensitivity test, which reassigns the uninsured ratio across banks and refits while holding the observed CAR vector, and with it the dependence structure, fixed. This is not design-based randomization inference. The ratios were not randomly assigned, so exchangeability under the null is an assumption, and reassigning the ratio breaks its correlation with size. What the test shows is how much of the estimate cross-sectional correlation alone could produce, which is exactly what HC3 does not address. Results appear in Section 5.3.

Cook's distance (Cook and Weisberg 1982), reported in Section 5.3, finds six observations above the conventional $4/n = 4/48 = 0.083$ threshold, of which SBNY ($D = 0.824$) and FRC ($D = 0.361$) are the most influential. That is what motivates the robustness check of Section 5.5, which re-estimates the cross-section without the three failed banks (SIVB, SBNY, FRC) and so on $N = 45$.

3.5 Fama-French three-factor robustness

As a robustness check on the market model, abnormal returns are also estimated with the Fama and French (1993) three-factor model, which adds two systematic factors to the market factor:

$$R_{it} - R_{ft} = \alpha_i + \beta_i(R_{mt} - R_{ft}) + s_i\text{SMB}_t + h_i\text{HML}_t + \varepsilon_{it} \tag{12}$$

Here R_{ft} is the risk-free rate (one-month U.S. Treasury bill rate), SMB_t is the size premium and HML_t the value premium. The loadings s_i and h_i are estimated jointly with α_i and β_i by OLS over the same window $[-210, -11]$, with daily factor returns from the Kenneth French Data Library. The check matters because regional bank stocks are mostly small- to mid-cap value stocks, so a single-factor model would push any size and value loading into the intercept and residuals. The comparison is reported in Section 5.5 and Appendix C, and the two sets of CARs correlate at 0.998.

3.6 Information dynamics channel

To look at the information environment around the observed CARs, Google Trends search volume is used as a proxy for public attention. Google Trends reports normalised search interest for a given term on a 0–100 scale, where 100 is the peak day of the series. Two terms are used, 'SVB' and 'bank run', and a composite index is built as their equally weighted average over

January to June 2023. Spikes in that index around the event dates are read as bursts of public attention of the kind Goldstein and Pauzner (2005) treat as a common public signal. Three limitations set the boundaries of what the series can carry. It is worldwide rather than specific to the United States, and it cannot be assigned to individual banks, which rules out a bank-level cross-sectional test. Attention also cannot be separated from the arrival of the crisis news itself, so co-movement between attention and returns says nothing about which one moved first. The series is therefore used to describe the pattern of public attention, not to establish a causal channel. These limitations are revisited in Section 5.4.

3.7 Identification, validity, and limitations

Several threats to the validity of the strategy need discussion. First, abnormal returns are measured against a market-model benchmark and not against a causal counterfactual. The identifying assumption, that returns would otherwise have followed the predicted path, cannot be tested, although it is standard in this literature.

A question comes before that one, namely whether a model of depositor withdrawal can be brought to equity-market data at all. The agents in Goldstein and Pauzner (2005) are depositors deciding whether to withdraw, while what is observed here is the repricing of bank equity, and no claim is made that the two face the same payoffs. What connects them is that the payoff structure of the model is not specific to deposit contracts. The authors extend it to “a run on a financial market”, in which investors rush to sell an asset and thereby cause its price to collapse, and observe that such settings display the same one-sided strategic complementarities (Goldstein and Pauzner 2005, 1311). Equity prices are read here in that spirit, as the market’s assessment of run risk, rather than as a direct measure of depositor behaviour. The direct test would require daily bank-level deposit flows, which are not published at that frequency.

A cleaner identification strategy is not available inside this episode. The obvious candidate is to treat the systemic-risk exception of March 12 as a difference-in-differences shock, under which high-uninsured banks should recover once run-risk is taken away. That does not work, because the exception protected only the depositors of SVB and Signature and left run-risk in place everywhere else. Consistent with this, the uninsured-deposit coefficient remains negative when the same cross-sectional specification is estimated on the Event 3 window rather than the Event 2 window (-0.0118, $t = -2.80$, $p = 0.008$, HC3, $N = 48$, $R^2 = 0.43$). That auxiliary regression is not otherwise used in this thesis and is reported here only to establish the point. With no exogenous variation in insurance coverage, the estimates are read as market-pricing relationships, and a design with genuine exogenous variation is left to future work.

Second, the five events are not independent of one another. They are stages of a single episode, and the $[-2,+2]$ windows of the four March events overlap on the calendar, so the same trading days enter more than one window and the CAAR series is positively autocorrelated across events. The five event-level CAARs therefore cannot be treated as independent observations, and they should be read together rather than as five separate tests. Event labels are retained only as a way of indexing windows. The calendar-time portfolio of Section 3.3 addresses this by placing all overlapping banks in a single daily portfolio, so cross-event correlation is absorbed into the portfolio return rather than biasing the standard errors.

A separate concern attaches to the attention panel of Section 5.4. Date fixed effects absorb everything common to a trading day, so the attention index enters only through its interaction with bank-level uninsured exposure. Identification then rests on the assumption that attention is the only day-varying factor that hits high-uninsured banks differently, and that assumption is a strong one. The severity of the crisis varied across exactly those days and is mechanically correlated with attention, because search volume peaked when the worst news arrived. The interaction therefore measures the joint effect of attention and contemporaneous news on the pricing of funding fragility, not an attention channel net of news. Separating the two would require attention data whose variation is predetermined relative to the crisis, for example the firm-level social-media exposure of Cookson et al. (2026), which a worldwide search series cannot provide.

Third, the cross-sectional regression of Section 3.4 suffers omitted variable bias if unobserved characteristics drive both balance-sheet structure and crisis-period returns. Banks with weaker loan portfolios may have both higher uninsured deposit ratios and more negative CARs, and that would bias γ_1. Tier 1 leverage and the loan-to-deposit ratio are only rough proxies for asset quality and capitalisation. Instrumental-variables or difference-in-differences designs that use exogenous variation in coverage are left to future work.

A related point concerns how many specifications are reported, which raises the question of which of them were planned and which followed a look at the data. The answer differs from one to the next. The confirmatory core was fixed in advance: the $[-2,+2]$ Event 2 CAR as dependent variable, the uninsured-deposit ratio as the variable of interest, and the three controls of Section 3.4. The calendar-time specification was also nominated before its result was known (Section 5.2). The rest is exploratory in the strict sense of having been chosen once the main estimates existed: the quantile regression, the progressive trimming, the quadratic and threshold tests, and the attention interaction. Those are the more interesting results, which is exactly why they are reported as descriptions of where the association lives rather than as tests it has passed. No p-value among them should be read as if it came from a pre-registered test.

Fourth, the cross-section covers 48 of the 54 banks in the initial target universe (Section 4.1). The six that were left out could not be included because complete and validated daily price series could not be assembled from the data providers used here. That is a limitation of data retrieval rather than an absence of trading history for those institutions. The six exclusions are not missing at random, since data availability rather than design decided them, although none of them was among the severely stressed institutions of March 2023. A coverage rate of 89% limits the scope for sample selection bias without removing it.

4 Data

4.1 Stock price data

Daily adjusted closing prices for 48 publicly traded U.S. bank stocks were taken from Yahoo Finance, with Stooq as a fallback, for April 1, 2022 to June 30, 2023, which is about 315 trading days. That is enough pre-event history for the $[-210, -11]$ estimation window of every event. The sample puts together the constituents of the KBW Nasdaq Regional Banking Index and the largest institutions from the KBW Nasdaq Bank Index, so it is a custom crisis sample that covers the whole size distribution of U.S. commercial banks rather than any single index. It includes the three subsequently failed banks (SIVB, SBNY, FRC), the two most severely stressed survivors (PACW, WAL), other stressed mid-size regionals (ZION, KEY, RF, CMA), large regionals and super-regionals (USB, TFC, FITB, HBAN, CFG), the three G-SIBs (JPM, BAC, WFC), and 31 smaller regional banks. This listing describes coverage of the size distribution and cuts across the three groups of Table 2, which are the definitions used in the analysis. Of an initial universe of 54, six are excluded because validated event-window price series could not be assembled (SNV, NYCB, HTLF, VBTX, FBMS, HTBK), which leaves 48. PacWest and Comerica are unavailable from the primary provider and are taken from secondary sources. PacWest comes from an archived series covering the period before its ticker was retired after the 2023 Banc of California merger. Comerica comes from a split-adjusted series. Both enter the event study, the daily-frequency analysis and the cross-section on the same footing as every other bank.

Two qualifications on coverage apply throughout. SVB and Signature ceased trading during the crisis, so the number of banks contributing to any given day falls after March 10 and is reported day by day in Table 4. Events 4 and 5 rest on the 46 institutions still listed at those dates. At Event 5, First Republic was seized before the open on day 0 and its shares were suspended for two sessions, resuming over the counter on May 3, so it contributes three of the five window days. It is retained on the same basis as the banks halted at Event 2. Its CAR is cumulated over the days it traded, the return across the suspension included, with the $\sqrt{n}$ scaling to match. And the six exclusions from the target universe are small or mid-size institutions, none severely stressed in March 2023, so the scope for selection bias is limited though not removed. Section 5.5 reports how the estimate responds to sample composition. Log returns are computed as $r_t = \ln(P_t/P_{t-1})$.

Some tickers were subsequently delisted (SIVB, SBNY, FRC, and PACW, the last after its late-2023 merger with Banc of California), and for these Yahoo Finance's bulk API no longer provides historical data. They are downloaded as CSV files from the Yahoo Finance website (`finance.yahoo.com/quote/TICKER/history`) covering the same date range, using the Adj Close column, which accounts for any splits and dividends.

4.2 Market index

The S&P 500 Composite Index (^GSPC) is the market portfolio in the market model, and it was downloaded from Yahoo Finance for the same date range. It is the standard benchmark in U.S. equity event studies and gives the broad market exposure needed to estimate systematic risk (beta).

4.3 Bank characteristics (FDIC Call Reports)

Bank-level balance sheet data are sourced from the FDIC BankFind API (`https://banks.data.fdic.gov/api/financials`) for the fiscal quarter ending December 31, 2022 (Q4 2022). That is the last reporting date before the crisis, so it shows the balance sheets that depositors and investors were working from when the crisis began.

Two features of this source create a mismatch with the dependent variable, and both of them probably bias the estimate. FDIC data are reported for the insured bank charter, while the CARs belong to the listed holding company. For single-charter regionals the two are the same thing. But the JPMorgan figure of $3,202bn is JPMorgan Chase Bank, N.A. rather than the group, and the same holds for U.S. Bancorp. The deposit denominator also includes foreign-office deposits, which are uninsured by construction. Both problems fall hardest on the largest internationally active banks and push their measured uninsured share up, which is largely why the 59.5 per cent of JPMorgan looks so high. Since those banks recorded the least negative CARs, this measurement error is correlated with the regressor and works against the hypothesis being tested, so the coefficient is more likely to be attenuated than inflated. Consolidated FR Y-9C data would remove the mismatch and would be the most direct improvement available.

Four variables are constructed. The Uninsured Deposit Percentage is (Total Deposits − FDIC-Insured Deposits) / Total Deposits × 100, using FDIC fields DEP and DEPINS. Total Assets in billions uses field ASSET. The Tier 1 Leverage Ratio (%) uses field RBC1AAJ. The Loan-to-Deposit Ratio (%) is Net Loans and Leases / Total Deposits × 100, using fields LNLSNET and DEP. Data for PACW were not available via the BankFind API and are sourced from PacWest Bancorp's Q4 2022 earnings release. PacWest reported an uninsured deposit share of 71.1 per cent, the highest in the sample outside the three failed institutions.

Table 2 summarises these four characteristics by group, with the bank-by-bank values for all 48 institutions in Table B.3. The corresponding Event 2 CARs are reported in Table C.4 (Appendix C). The summary already shows the variation that the cross-section works with. The three failed banks average 83.6 per cent uninsured funding against 42.4 per cent among the regionals, while being an order of magnitude smaller than the large banks in assets. The spread is wide. Uninsured deposit percentages run from 32.4% (Bank OZK) to 94.3% (Silicon Valley Bank), a range of more than 60 percentage points. PacWest (71.1%) is third-highest in the sample and the highest of the banks that did not fail, and Comerica (59.3%) is sixth (see Appendix B, Table B.3). That variation is exactly what the cross-sectional regression relies on.

Group	N	Uninsured % (mean)	Uninsured % (range)	Assets $B (mean)	Tier 1 % (mean)	LTD % (mean)
Failed	3	83.6	67.3–94.3	177.3	8.42	73.3
Large	5	46.9	39.3–59.5	1693.8	8.18	61.5
Regional	40	42.4	32.4–71.1	55.3	9.12	78.4
All banks	48	45.4	32.4–94.3	233.6	8.98	76.3

Table 2: Bank characteristics by group (Q4 2022), $N = 48$. Note: the groups are Failed (SIVB, SBNY, FRC), Large (assets > $500B) and Regional (others). These three labels are used consistently throughout the thesis. Uninsured % = uninsured deposits / total deposits. Assets $B = total assets in billions. Tier 1 % = Tier 1 leverage ratio. LTD % = loan-to-deposit ratio. Bank-by-bank values for all 48 institutions are given in Table B.3. Source: FDIC BankFind Q4 2022 and author's calculations.

4.4 Google Trends data

Daily search-volume indices for two keywords, 'SVB' and 'bank run', are obtained from Google Trends (`trends.google.com`), worldwide, over the period January 1 to June 30,

2023. The series was put together from two overlapping three-month downloads and rescaled onto a common basis using the days they share. A single January–June extract returns daily data directly and would have given the same series up to a constant, so nothing that follows depends on the splicing, which is described here only for completeness. The data are relative search interest, normalised to 0–100, where 100 is the day of peak interest. A composite information index is built as the average of the two normalised series.

The composite index is rescaled so that its own peak equals 100, and that peak falls on March 13, 2023. On that day the 'SVB' series stands at 100 and 'bank run' far lower, so the unscaled average of the two is well below 100, at 53.5. How the index behaves around the event dates is described in Section 5.4.

4.5 Descriptive statistics

Table C.1 (Appendix C) shows descriptive statistics for the estimation-window stock return distributions and the market model estimates. It covers a representative subset of fourteen banks across the size and volatility range of the sample. The R-squared values are moderate, running from 0.25 to 0.64 across the 48 banks and from 0.37 to 0.64 among the fourteen shown. Bank stocks move with the market a good deal in normal times, but most of their daily variation stays idiosyncratic, and that is what makes the market model informative here. Summary statistics for the four balance-sheet variables and for the Event 2 CARs across the full 48-bank cross-section are given in Table B.1 (Appendix B), and the correlations among them in Table B.2.

5 Empirical results

5.1 Event study results: CARs at each event

Event 2, the window centred on the SVB closure of March 10, 2023, is the primary window of interest, and Table C.4 in Appendix C lists the CAR of every bank over it. One qualification applies to every event-level figure in this section. The $[-2,+2]$ windows of the four March events overlap on the calendar. The Event 2 window runs from March 8 to March 14 and so contains both the March 8 disclosure and the March 13 session on which the systemic-risk intervention was first priced. What the CAR measures is therefore the reaction to the whole

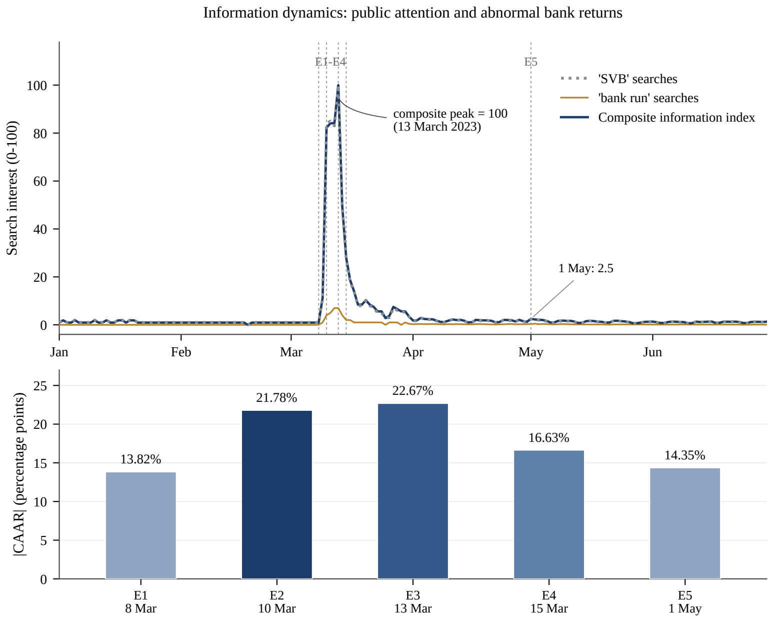

Figure 1: Google search interest for "SVB" and "bank run" and the composite information index (top panel), and the absolute Cumulative Average Abnormal Return by event (bottom panel), January–June 2023. Top panel: daily Google Trends indices, worldwide. The heavy line is the composite index actually used in Section 5.4, rescaled so that its own peak equals 100 on March 13. It tracks the 'SVB' series closely because 'bank run' search volume is more than an order of magnitude smaller throughout. Bottom panel: |CAAR| in percentage points per event, on the bank-level mean of Table 3. Dashed vertical lines in the top panel mark the five event dates. Source: Google Trends and author's calculations.

March cluster as seen in that window, and not the effect of the closure on its own. The CAAR point estimates for all five windows are reported in Table 3. Every figure there is in log-return units and every one is computed the same way, as the cross-sectional mean of bank-level CARs over the window. Events 4 and 5 rest on 46 banks rather than 48, because SVB and Signature had ceased trading by then.

One result needs a comment straight away. Event 3 gives the most negative CAAR of the five on the bank-level mean used throughout this section, although by less than a percentage point over Event 2. The ranking holds under the other aggregation as well, since -22.67% lies below both of the Event 2 figures reconciled in Section 5.2, the bank-level mean of -21.78% and the day-summed -22.31%. That is the opposite of what Section 2.5 leads one to expect, since the systemic backstop should have been received as good news. However, that expectation runs two different groups of claimants together, and the puzzle disappears once they are separated.

The systemic risk exception was an intervention on behalf of depositors and not of shareholders, and for equity holders it carried three negative signals. Invoking an emergency author-

Event	Announcement	Day 0	Event window	N	CAAR (%)
1	Wed, 8 Mar	8 Mar	6 – 10 Mar	48	−13.82
2	Fri, 10 Mar	10 Mar	8 – 14 Mar	48	−21.78
3	Sun, 12 Mar	13 Mar	9 – 15 Mar	48	−22.67
4	Wed, 15 Mar	15 Mar	13 – 17 Mar	46	−16.63
5	Mon, 1 May	1 May	27 Apr – 3 May	46	−14.35

Table 3: Cumulative Average Abnormal Returns by event, $[-2,+2]$ window. Note: every CAAR here is the cross-sectional mean of bank-level CARs in log-return units, the aggregation used throughout Sections 5.1 and 5.3 and plotted in Figure 1. It is not the day-summed CAAR of Table 4, which at Event 2 gives −22.31% against −21.78% here, reconciled in Section 5.2. All dates are 2023, and the announcement and day-0 columns are those of Table 1. They coincide except at Event 3, whose announcement fell on a Sunday, so day 0 is the next trading session and the window is dated from it. Windows are the calendar span of the five trading days around day 0. N counts the banks entering each mean, a bank entering if it traded on any day of the window (Section 4.1). The four March windows overlap, so these figures are not independent observations (Section 3.7). Source: author's calculations.

ity reserved for systemic threats told the market publicly that the situation was bad enough to warrant it. The way depositors were made whole was FDIC receivership, in which shareholders are wiped out, so a policy that guarantees deposits at failed institutions implies zero equity recovery for any bank that goes down the same road, and it raises the perceived probability of that road for the others. And protecting uninsured depositors after the fact does nothing to restore the deposit franchise value that leaves with them, which, as Drechsler et al. (2026) emphasise, is itself the runnable asset. Read this way, the negative CAAR at Event 3 shows that the intervention was informative rather than economically harmful, because it confirmed how serious things were while offering shareholders nothing. The overlap between the Event 2 and Event 3 windows (Section 3.7) means part of the Monday March 13 decline is continued pricing of the closures rather than a response to the exception itself, but the sign is unambiguous and the equity-holder logic explains it without appeal to market irrationality.

Three patterns are immediately apparent. First, all 48 banks experience negative CARs at Event 2. Second, the magnitude falls along a steep gradient. It runs from the failed banks (FRC −104.5%, SIVB −89.0%, SBNY −33.8% in log units) and the most stressed regionals, Western Alliance (−85.3%) and PacWest (−69.1%), through the mid-size regionals (CMA −38.9%, ZION −37.4%, KEY −34.7%, RF −14.4%) and large super-regionals (TFC −30.8%, USB −18.3%) to the G-SIBs (BAC −12.0%, WFC −8.8%), with JPMorgan (−1.9%) alone largely spared. Third, this gradient shows that the market told banks apart sharply instead of marking the whole sector down evenly, which does not fit a purely indiscriminate panic. It does not, however, settle the choice between the two accounts of Section 2.1. Uninsured-deposit concentration measures run-vulnerability and correlates with general balance-sheet fragility at the same time, and both accounts predict the ordering that is observed. What the gradient does establish on its own terms is that equity valuations separated banks systematically within days, and any account of the crisis has to explain that.

Turning from size to statistical significance, the reaction was severe across most of the sample but not everywhere. The failed and near-failed institutions (SIVB, FRC, SBNY, WAL) and the mid-size and large banks (ZION, KEY, TFC, USB, BAC, RF) are all significant at the 0.1% level, and Wells Fargo at the 1% level. Significance weakens steadily as the CAR shrinks. With residual standard deviations of 0.0069 to 0.0339 (Table C.1), a five-day CAR must reach roughly $1.96 \times \hat{\sigma}_i \times \sqrt{5}$ to clear the 5% level, which runs from about 3 percentage points for the least volatile banks in the sample to about 15 for the most volatile. Whether a given bank in Table C.4 is individually distinguishable from zero therefore depends on its own estimation-window volatility as well as on the size of its CAR. JPMorgan (-1.90%, $t = -0.74$, $p = 0.460$) is the clearest such case and the only one discussed here, since it is the only bank whose small reaction carries an economic interpretation. (The per-bank significance among the more affected institutions is not in tension with the aggregate result of Section 5.2. Each such bank's abnormal return is large relative to its own normal volatility. But the banks all moved together, so the sector-wide average is not distinguishable from zero once that cross-sectional dependence is netted out.)

All CARs are cumulative log-return abnormal returns over the $[-2,+2]$ window, so a value below -100% is admissible and does not violate limited liability. The t-statistic is $\mathrm{CAR}/(\hat{\sigma}\sqrt{n})$, where n is the number of event-window days on which the bank traded. Because n is bank-specific, CARs are not summed over an identical window for every institution. SIVB was halted on the morning of March 10 and last traded on March 9, and SBNY last traded on March 10, so each contributes fewer than five daily abnormal returns while surviving banks contribute the full five. The scaling by $\sqrt{n}$ keeps the t-statistic comparable, but the CAR level for a halted bank is cumulated over a shorter interval and is, if anything, understated relative to a survivor's. The failed banks drive the tail concentration documented in Section 5.5, so two checks reported there address this directly.

5.2 AAR and CAAR analysis

Table 4 presents the day-by-day Average Abnormal Returns (AAR) and Cumulative Average Abnormal Returns (CAAR) for Event 2 across all 48 banks. The pattern begins with pressure on day -2 (-0.57%***) and then a collapse on day -1 (-7.48%***), the first full trading session after the disclosure by SVB on the evening of March 8. That fits MacKinlay (1997, Section 2), who justifies wider event windows in order to catch information leakage around announcements. A further fall follows on day 0 (-2.99%*). The steepest single-day decline comes on day $+1$ (-13.51%***), which is Monday March 13, when markets opened after the weekend in spite of the backstop announced the previous evening. That the depositor backstop did not stop the fall in equity fits the reading of Section 5.1, since the intervention protected depositors and not shareholders. By day $+2$ the CAAR had reached -22.31%, a total market-adjusted loss of more than 22 percentage points for the average bank in the sample. The partial recovery on day $+2$ ($+2.23\%$*) fits markets starting to settle once the systemic risk exception had been absorbed.

Two aggregates appear in this thesis and should not be confused. The -22.31% reported here sums the five daily cross-sectional means of Table 4, each taken over the banks trading that day. The -21.78% of Table 3 is the mean of the 48 bank-level CARs. They differ by 0.53 percentage points for one reason only, that the sample shrinks from 48 banks to 46 during the window as SVB and Signature stop trading. Neither is more correct. The bank-level mean is used wherever the cross-section is the object of interest, since it is the quantity the regressions of Section 5.3 explain, and the naive cross-sectional t-statistic below is computed from it.

The full Patell (1976) standardised-residual test gives large aggregate Z-statistics (Event 2: $Z = -47.2$), which come from the sheer scale of the failed-bank abnormal returns relative to their estimation-window volatility. These values have to be read with care, because the Patell test assumes the banks are independent, and that assumption fails badly here. All 48 of them react to the same events on the same calendar days, with an average pairwise abnormal-return correlation of 0.616. Two standard corrections are applied. They reach different conclusions, and both are reported.

The first is the cross-sectional-dependence adjustment of Kolari and Pynnönen (2010), which inflates the variance of the Event 2 cross-section by the average pairwise correlation. The adjusted statistic is $t\sqrt{(1-\bar{r})/(1+(n-1)\bar{r})}$. The numerator corrects for the downward bias in the cross-sectional sample variance under correlation, and the denominator for the inflation of the variance of the mean. With $\bar{r} = 0.616$ and $n = 48$ the factor is 0.113. That brings the naive cross-sectional t down from -6.82 to -0.77, which does not reject the null of zero aggregate abnormal return. This correction works directly on the dependence structure the design actually has, and it uses the full 48-bank cross-section rather than a short time series. It is therefore treated here as the more demanding and more conservative of the two, and the aggregate conclusion is stated on its terms.

The second is the calendar-time portfolio regression described in Section 3.3. On each calendar day inside at least one event window, the banks in that window form a single equally

Day	AAR	t(AAR)	p(AAR)	CAAR	N Banks
-2	-0.57%***	-4.48	< 0.001	-0.57%	48
-1	-7.48%***	-4.09	< 0.001	-8.05%	48
0	-2.99%*	-2.52	0.015	-11.04%	47
$+1$	-13.51%***	-5.41	< 0.001	-24.55%	46
$+2$	$+2.23\%$*	$+2.59$	0.013	-22.31%	46

Table 4: Average Abnormal Returns (AAR) and Cumulative Average Abnormal Returns (CAAR), Event 2. Note: AAR = equally weighted average daily abnormal return. CAAR = its running sum, accumulated from unrounded values, so the rounded AARs shown sum to -22.32% rather than the -22.31% printed. N_τ varies by day, because SIVB last traded March 9 and SBNY March 10, while First Republic traded on all five. t = AAR / SE(AAR) across the banks trading that day, and the p-column and stars refer to that test alone. The CAAR column carries no marks, since the test is of the day's average rather than of the running total. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, $\cdot$ $p < 0.10$ (two-tailed). These day-level stars come from tests treating banks as independent, which Section 5.2 shows they are not, so they are not evidence of a sector-wide effect. Day 0 = March 10, 2023 and day $+1$ = March 13. Source: author calculations.

weighted portfolio, whose return is regressed on the market return over the fifteen crisis days with Newey-West standard errors. The intercept, which estimates the average daily abnormal return, is -2.71% per day with $t = -3.86$ ($p = 0.002$). The market beta of 2.83 should not be taken as confirmation of anything, because it is estimated on the fifteen crisis days themselves rather than over a long series that also contains non-event days, as the standard implementation asks for. Set against the estimation-window betas of 0.78 to 1.77 in Table C.1, and 0.27 to 1.77 across all 48 banks, a loading of 2.83 is better read as a symptom of the event contaminating the factor loading than as a finding in its own right. It also means the intercept is measured against a beta that the event itself helped to set. Cumulated over five days this implies a CAAR of -13.56%, with a 95% interval of $[-20.44\%, -6.68\%]$ computed on the normal approximation. The t critical value appropriate to the thirteen residual degrees of freedom widens it to $[-21.14\%, -5.97\%]$, and that wider interval is the one reported. Two things limit the weight this estimate can carry. It rests on only fifteen time-series observations, which is a thin basis for inference, and it pools all five event windows instead of isolating any one of them. It is therefore reported as a complementary specification and not as the basis for the aggregate conclusion.

Three things should be taken from this section. The two procedures disagree, and that disagreement is not resolved here. The dependence-adjusted test cannot show the sector-wide average to differ from zero, while the portfolio regression rejects, and the first is treated as primary for the reasons given above, so the aggregate conclusion of this thesis is a failure to reject rather than a demonstration that nothing happened. Both agree that naive inference overstates the evidence, but only one speaks to magnitude: the Kolari-Pynnönen adjustment rescales the test statistic and leaves the -21.78% point estimate untouched, while the portfolio regression supplies a second estimate, roughly -14% over five days against the naive -22%. The gap is not explained by the failed banks, since excluding all three moves the cross-sectional mean only from -21.78% to -18.18%. It reflects the different way each procedure handles the component common to every bank. Neither point weakens what follows, because the cross-sectional question of which banks fell furthest does not depend on whether the average fell at all. The informative question is why some institutions were repriced far more severely than others, to which the analysis now turns.

The cross-sectional pattern can be stated with more confidence, because it does not depend on the aggregate test. The spread of CARs is wide and ordered in a systematic way. Four tail cases, the three failures plus the near-failure Western Alliance, contribute more than their share of the sector average, though not most of it. They come to -312.6 of the -1045.3 percentage points summed across all 48 banks, or 29.9 per cent of the total, and taking them out still leaves the average bank down 16.7 per cent. The more telling number is the gap between mean and median. The median bank lost 14.5 per cent against a mean of 21.8 per cent, so the distribution is strongly skewed by its lower tail even though that tail is not where most of the average comes from. The dominant feature of the data is therefore concentration rather than uniformity, and it is that concentration, quantified in Section 5.5, on which the interpretation rests. Global-games models (Goldstein and Pauzner 2005) associate such a pattern with institutions near a fragility boundary.

5.3 Cross-sectional regression results

Table 5 reports the cross-sectional regression of the Event 2 CARs on bank characteristics for all 48 banks. Two specifications are estimated: the full model, which uses four explanatory variables, and the parsimonious model, which keeps only the two variables the theory points to.

The main cross-sectional result concerns the uninsured deposit ratio. In the parsimonious model the coefficient is -0.0112 and it is significant at the 1% level ($t = -3.05$, $p = 0.004$). In the full model, which is the preferred specification, it is -0.0115 ($t = -2.53$, $p = 0.015$). Banks with a higher share of uninsured deposits lost more value, which is what one expects if equity returns move with a bank's exposure to information-driven run risk. The dependent variable is the CAR in decimal form and the uninsured ratio is measured in percentage points. This means that a 10 percentage-point increase in the uninsured deposit ratio goes together with a CAR that is about 11.2 percentage points more negative in the parsimonious specification and 11.5 in the full one. The 95% confidence interval of the parsimonious estimate is $[-0.0185, -0.0038]$, so it lies well away from zero. HC3 standard errors do not deal with the cross-sectional dependence reported in Section 5.2, so the coefficient was also put through the permutation test described in Section 3.4. The uninsured-deposit ratio was reassigned across banks 20,000 times while the observed CAR vector was held fixed, and in neither specification did any permutation produce a coefficient as large in absolute value as the observed one, so the two-sided p-value is below 0.0001 in both cases. These figures are much smaller than the HC3 p-values, but this is a property of the two procedures rather than extra evidence, and the HC3 figures remain the ones reported throughout. The test has two limits. Uninsured ratios were not randomly assigned, so exchangeability is assumed rather than given, and reassigning the ratio on its own breaks its correlation of 0.36 with log(assets) (Table B.2), which a residual scheme of the Freedman-Lane type would have preserved. The exercise is therefore a sensitivity check against cross-sectional correlation, not a permission to read the association as causal. Figure 2 plots the scatter behind the regression together with the parsimonious fit, and it makes the shape of the association visible. Most of the slope comes from the lower part of the return distribution, and Section 5.5 measures that heterogeneity directly. The two specifications agree with each other: the association is significant at the usual levels whether or not the extra balance-sheet controls are included.

The log(Total Assets) coefficient is close to zero and insignificant in both models (0.0011, $t = 0.050$ in the parsimonious model and 0.0113, $t = 0.506$ in the full one). Size is therefore not a useful predictor of the cross-sectional variation in CARs once the uninsured deposit ratio is in the regression. The Tier 1 leverage and loan-to-deposit ratios are insignificant in the full model as well. The parsimonious model explains 44.8% of the variation in Event 2 CARs and the full model 53.1%, so observable balance-sheet characteristics, and above all the uninsured deposit ratio, account for roughly half of the differences in how banks were repriced. The rest comes from things the model does not contain, such as bank-specific social-media exposure (Cookson et al. 2026), interest-rate hedging and the composition of the deposit base. Two remarks belong here. With $N = 48$ and four regressors the unadjusted R^2 is biased upwards, so the adjusted R^2 is the better summary: it is 0.487 for the full model against 0.424 for the parsimonious one. The full model is used as the preferred one throughout, but the reason is that Tier 1 leverage and the loan-to-deposit ratio are theoretically relevant confounders, and not that comparison of fit, which would be a weak basis for choosing a specification. The parsimonious model is kept because it is the one comparable to the earlier literature, and the estimate hardly moves between the two, -0.0115 against -0.0112, with both significant at the 5% level or better. These cross-sectional R^2 values should not be confused with the time-series R^2 of the market-

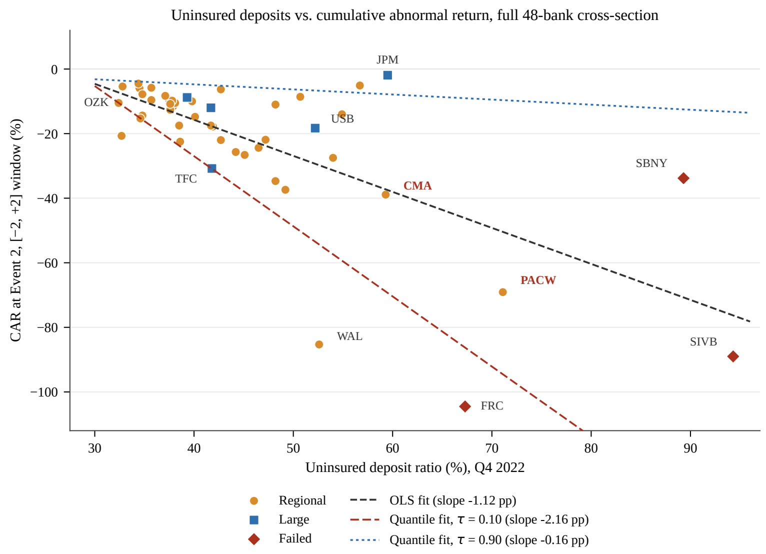

Figure 2: Uninsured deposit ratio versus cumulative abnormal return over the Event 2 window, full 48-bank cross-section. Note: each point is one sample bank, and colours denote the groups defined in Table 2. The dashed line is the OLS fit through all 48 observations. Its slope of -1.12 percentage points is the parsimonious coefficient of -0.0112 from Table 5 on the percentage-point scale. The two additional lines are the corresponding quantile-regression fits at $\tau = 0.10$ and $\tau = 0.90$ from Table 7, evaluated at the sample mean of $\log(\text{assets})$. The fan between them is the heterogeneity the quantile estimates measure, and the reason the relationship cannot be summarised by the OLS slope alone. Labels are shown for the most-discussed institutions only. Source: author's calculations from Tables 2 and C.4.

model estimations in Table C.1 (0.25–0.64 across the sample), which measure how strongly a bank co-moves with the market and not how well the model explains the cross-section of CARs.

A supplementary regulatory specification regresses the CAR on the uninsured ratio, $\log(\text{assets})$, the EGRRCPA exemption dummy and a G-SIB indicator, which equals 1 for JPM, BAC and WFC, each of them above \$700 billion in assets at Q4 2022. The G-SIB coefficient is $+0.328$ ($t = +2.08$, $p = 0.044$), and on the decimal CAR scale of Table 5 this is a premium of about 33 percentage points relative to otherwise similar banks. The uninsured deposit coefficient stays significant (-0.0098, $t = -2.70$, $p = 0.010$), while $\log(\text{Assets})$ and the exemption dummy are both insignificant. The exemption dummy in particular carries a coefficient close to zero and is reported for completeness rather than because it separates banks from one another. This specification has to be read with care, but the reason is the size of the treated group and not collinearity. The G-SIB indicator is switched on for exactly three banks, so the premium is a comparison between those three institutions and the remaining 45, and it cannot be attributed to the G-SIB designation as such rather than to sheer size or to anything else those three banks have in common. The variance inflation factors in Table C.2 show that the specification is still estimable: in the widest model log(assets) reaches 6.22 and the G-SIB dummy 2.65, while the uninsured deposit ratio is almost untouched at 1.38. Section 5.5 shows that dropping the three failed banks leaves the uninsured coefficient essentially where it is.

Before the OLS estimates are interpreted, four standard diagnostic checks were run, and they are reported in full in Appendix C (Tables C.2 and C.3). Multicollinearity is moderate, on the variance inflation factors already given. Homoskedasticity is rejected (Breusch-Pagan LM $= 21.16$ on six degrees of freedom, $p = 0.002$), which shows that HC3 standard errors are necessary here and not merely a precaution. Normality of the residuals is rejected as well (Jarque-Bera $= 14.20$, $p < 0.001$), and this is driven by the very large negative CARs of SIVB and FRC. The HC3 sandwich estimator does not require normally distributed residuals, although with $N = 48$ its inference remains approximate. Cook's distance flags six observations above the $4/n = 4/48 = 0.083$ threshold, and SBNY is by far the most influential one ($D = 0.824$). That influence sits in the failed banks, which is exactly why the exclusions and the resampling checks of Section 5.5 are the right response.

Variable	Full Coef.	Full t	Full p	Pars. Coef.	Pars. t	Pars. p
Intercept	0.006	0.010	0.992	0.284	1.967	0.055
Uninsured Deposits (%)	−0.0115	−2.525	0.015*	−0.0112	−3.051	0.004**
log(Total Assets)	0.0113	0.506	0.615	0.0011	0.050	0.960
Tier 1 Leverage (%)	0.0609	1.157	0.254	—	—	—
Loan-to-Deposit (%)	−0.0039	−1.464	0.151	—	—	—
R-squared		0.531			0.448	
Adjusted R-squared		0.487			0.424	
N (banks)		48			48	
Robust SEs (HC3)?		Yes			Yes	

Table 5: Cross-Sectional OLS Regression. Dependent Variable: CAR at Event 2. Note: Dependent variable is the CAR over the $[-2,+2]$ event window around March 10, 2023, in decimal form. $N = 48$ (the custom KBW sample of Section 4.1, from which 6 tickers with unavailable price series are excluded). Standard errors are HC3 robust. The intercept is not interpreted in either specification: OLS fits the sample means exactly, so it is determined by the slopes and the regressor means. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, · $p < 0.10$. Source: author calculations using FDIC BankFind Q4 2022 data and event study CARs.

5.4 Information dynamics

This section reports stylised facts about attention during the crisis. It is descriptive evidence in support of the information-dynamics reading, not a test of a causal mechanism (Section 3.7). The daily Google Trends series (worldwide, January–June 2023) shows that public search interest in 'SVB' was almost nothing before the crisis, at a normalised value of 1–2 through January and February. It then jumped very sharply and reached its maximum of 100 on March 13, 2023, which is the trading day of the strongest market reaction (Event 2, day +1). Search interest in 'bank run' rose at the same time, but much less, with a peak of 7. Interest then fell away quickly during April. The First Republic seizure on May 1 (Event 5) produced only a small second bump, a normalised value of about 2.5 on the composite scale, which is roughly forty times smaller than the SVB peak. This almost vertical spike in the days around the closure of SVB is the visible signature of the information-driven dynamics of Goldstein and Pauzner (2005), and it is shown in Figure 1.

The search series is worldwide and not bank-specific, so it cannot carry a cross-sectional test linking the attention paid to an individual bank to that bank's returns. The evidence is limited to the co-movement between the timing of the search spike and the timing of the largest abnormal returns, which fall in the same few days of March 2023. Reverse causality makes it weaker still. Falling prices probably drive search just as much as attention drives prices, and the bidirectional lead-lag relationships documented by Bales and Burghof (2024) suggest that the two worked together, so no direction of influence is read out of the co-movement. A stronger test would need bank-specific attention data, such as the firm-level social-media counts of Cookson et al. (2026), and this is left to future work.

Seen through global-games theory, the search series is best read as a real-time proxy for the common public signal that coordinates the behaviour of depositors and investors. The near-vertical spike of March 8–13 marks the moment when idiosyncratic balance-sheet vulnerabilities, which were privately held information, met synchronised public attention, and the spike and the largest abnormal returns fall inside the same trading week. That co-timing is the observation. Whether attention turned those vulnerabilities into repricing is what the design cannot establish, because severe news arriving on the same days would look exactly the same.

This co-timing can be examined with a two-way fixed-effects panel, which goes beyond the single-crisis cross-section. The four March windows overlap, so pooling the five event windows without any adjustment would enter the same bank-day into the panel once for every window that contains it. A bank's abnormal return on a given date is one single quantity, so this repetition adds no information. It only re-weights the estimator towards the most overlapped dates and makes the sample look larger than it is. The panel is therefore collapsed to one observation per unique bank-date, which removes 468 repeated rows and leaves 697 bank-day observations across all 48 banks and 15 trading dates. The specification interacts each bank's time-invariant uninsured-deposit ratio with the time-varying information index and absorbs bank and date fixed effects:

$$\text{AR}_{it} = \gamma(\text{UninsuredDep}_i \times \text{Info}_t) + \text{bank FE} + \text{date FE} + u_{it} \tag{13}$$

The estimates are in Table 6: $\gamma = -0.0044$ ($t = -2.06$) with the uninsured interaction alone, -0.0045 ($t = -2.24$) when the size interaction is added, and -0.0045 ($t = -1.53$) when the regulatory-exemption interaction is added as well. The sign is the one the information-dynamics reading predicts and the point estimate is stable across all three specifications. However, none of them reaches conventional significance once the cluster structure is taken seriously. Inference needs particular care here. With only fifteen date clusters the asymptotic cluster-robust p-values over-reject by a wide margin (Cameron, Gelbach, and Miller 2008), so they are not the basis of any claim: they are 0.059, 0.042 and 0.149, while a wild cluster bootstrap that imposes the null and resamples by date gives 0.306, 0.304 and 0.352, on a within-R^2 of 0.018 in all three cases. The honest reading is that the panel cannot separate the interaction from zero. In any case, two things limited what it could have shown. Fifteen clusters sits at the lower bound of what the bootstrap handles reliably, so the test has little power. And date fixed effects absorb everything that is common to a trading day, so attention can enter only through its interaction with bank-level exposure, and that cannot be separated from the severity of the news arriving on the same day (Section 3.7). What survives is the descriptive statement already made above: search interest and the largest abnormal returns peak in the same few days. The panel does not turn this into a measurable difference across banks, and that null is reported here rather than left out.

Variable	Model 1	Model 2	Model 3
Uninsured × Info	−0.0044 (−2.06)	−0.0045 (−2.24)	−0.0045 (−1.53)
Bootstrap p (by date)	0.306	0.304	0.352
log(Assets) × Info	—	0.0030 (0.57)	0.0025 (0.42)
RegExempt × Info	—	—	−0.0021 (−0.53)
Bank fixed effects	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes
Within R^2	0.018	0.018	0.018
Observations	697	697	697

Table 6: Panel fixed-effects: the uninsured-deposit effect on high-attention days (dependent variable: daily abnormal return, $N = 697$). Note: two-way fixed effects (bank and date), with t-statistics from two-way clustered standard errors in parentheses. Inference rests on the wild cluster bootstrap in the second row, which imposes the null and resamples by date with Rademacher weights ($B = 999$). Info = worldwide Google Trends composite ('SVB' + 'bank run'), rescaled to the unit interval. Source: author's calculations.

The limitations of Section 3.6 shape how this should be read. The decisive one is that attention cannot be separated from the arrival of crisis news, so the interaction picks up the joint effect of attention and contemporaneous severity. The result is a stylised fact about when the cross-sectional association was strongest, and not evidence that attention caused it. The fifteen dates are also consecutive days inside a single crisis rather than independent draws. The conclusion is therefore a modest one. Attention and the uninsured-deposit penalty move together, which is consistent with an information-amplification channel but does not establish one. RQ3 was posed descriptively in Section 1.2 and it is answered descriptively here.

5.5 Robustness checks

Several additional analyses were run to check that the results hold up, and the routine ones are reported in full in Appendix C. Re-estimating the market model on a narrower $[-110, -11]$ window, and replacing the S&P 500 with a value-weighted bank index, leave the pattern of signs and significance unchanged.

Excluding the three failed banks ($N = 45$) leaves the uninsured-deposit coefficient almost unchanged and still significant ($\beta = -0.0103$, $t = -2.36$, $p = 0.023$). This is the single most important result in this section, and it follows directly from the way the sample was built in Section 4.1. Two banks were repriced severely without failing, so the surviving-bank subsample still contains institutions that combined a high share of uninsured funding with large losses. The association therefore cannot be dismissed as an artefact of the three banks that collapsed. This exclusion rejects at the usual levels rather than failing to reject, so no argument about statistical power is needed to interpret it: the surviving-bank subsample is not being asked to prove a null. A bank-level bootstrap with 2,000 resamples gives a mean coefficient of -0.0113 and a 95% percentile interval of $[-0.0180, -0.0050]$. Leave-one-bank-out keeps the coefficient negative in all 48 re-estimations, in a range from -0.0096 to -0.0140, with Signature again the most influential single observation. A Huber M-estimator gives -0.0118, slightly larger in absolute value than the OLS estimate, which shows that the coefficient is not held up by a few extreme observations.

Two further checks are reported in Appendix C. Direct parametric tests of the threshold interpretation, a quadratic term and an interaction with an indicator for uninsured deposits above 60%, are both null. However, only four banks lie above that cut-off, so these tests have very little power and their failure to reject is not evidence against non-linearity. The second check concerns the two banks whose CARs are cumulated over shortened windows, and they are not what produces the result: restricting the sample to the complete-window banks, and rescaling to a mean daily abnormal return, both leave the coefficient significant and larger in absolute value.

Quantile regression re-estimates the parsimonious specification across the conditional CAR distribution instead of at its mean, and the pattern is monotonic and economically stark. At the 10th percentile the uninsured-deposit coefficient is -0.0216, nearly twice the OLS mean estimate. It falls to -0.0124 at the median and to -0.0016 at the 90th percentile, where it cannot be told apart from zero. The banks whose outcomes were worst were therefore penalised for uninsured funding at a rate an order of magnitude higher than the banks that came through the crisis well. There are two limits, and they should be stated plainly. The lower-tail estimates are identified largely from the same extreme observations that drive the OLS coefficient, so this describes where the association sits rather than confirming it independently. And quantile regression tests a different hypothesis from the quadratic and threshold specifications, because it characterises the slope at points of the conditional distribution and not a cut-off in the regressor, so it cannot deliver the threshold evidence those tests failed to provide. With those qualifications in mind, the shape of the estimates is the shape that the three-region structure of Goldstein and Pauzner (2005) implies: steep near the boundary of the region in which fundamentals select the outcome, and flat well above it. The pattern is consistent with that structure, but it does not establish it.

One further qualification matters. When the three failed banks are excluded, the tail gradient becomes weaker and loses significance ($\tau = 0.10$: $\beta = -0.0144$, $t = -1.45$). The quantile evidence therefore shows that the effect is concentrated in the tail of the full sample, but not that a comparable gradient survives among the surviving banks on their own.

The standard errors, t-ratios and confidence intervals behind these point estimates are reported in Table 7.

Quantile (τ)	β (Uninsured Dep.)	Bootstrap SE	t	95% CI
0.10	-0.0216	0.0082	-2.62	$[-0.0377, -0.0055]$
0.25	-0.0125	0.0045	-2.75	$[-0.0213, -0.0037]$
0.50	-0.0124	0.0042	-2.95	$[-0.0206, -0.0042]$
0.75	-0.0048	0.0034	-1.39	$[-0.0115, +0.0019]$
0.90	-0.0016	0.0027	-0.59	$[-0.0069, +0.0037]$
OLS (mean)	-0.0112	0.0037	-3.05	—

Table 7: Quantile regression of Event 2 CAR on the uninsured deposit ratio and log(assets), $N = 48$. Note: standard errors come from 400 bank-level bootstrap resamples and the confidence intervals are the normal approximation $\hat{\beta} \pm 1.96$ SE on them. The t-ratios use the unrounded standard errors, so they differ in the second decimal from $\hat{\beta}$ divided by the four-decimal figures shown. Estimated by the Barrodale–Roberts simplex algorithm, the quantreg default. The OLS row is reproduced from Table 5. Source: author's calculations.

To see whether the association is about failure in particular or about extreme outcomes in general, the sample is trimmed step by step from the lower tail. Table C.6 reports the sequence.

Removing the three institutions that failed leaves -0.0103 ($p = 0.023$), and removing the worst-affected fifth leaves -0.0002 ($p = 0.911$). One step in Table C.6 looks strange at first sight and would otherwise raise the suspicion of an error: the coefficient falls in absolute value from -0.0064 to -0.0039 while its p-value falls from 0.064 to 0.004. Both move because the bank removed at that step, PacWest, is by a clear margin the most influential observation in the subsample it is dropped from, with a Cook's distance there of 1.04. Dropping it shrinks the estimate and the standard error at the same time, and the standard error by proportionally more. The significance at that step is therefore evidence of a weaker relationship estimated more precisely, not of a stronger one. Two things follow. The association is not a property of the failed institutions, because it survives their removal at conventional significance. But it is clearly concentrated in the lower part of the return distribution: among the thirty-eight banks outside the worst-affected fifth it cannot be told apart from zero. What identifies the relationship is therefore the group of banks that were severely repriced, whether or not they failed in the end, and PacWest is the clearest case of this.

Finally, a Fama-French three-factor check on all 48 banks confirms that the CARs are not an artefact of the single-factor benchmark. The two series correlate at 0.998 and keep the broad ordering of banks, and the reshuffling that does occur is limited to small changes in rank. Re-estimating the parsimonious regression on the FF3 CARs gives a coefficient that cannot be told apart from the market-model estimate (-0.0116, $p = 0.003$). Table C.5 sets out that comparison, with further detail in Appendix C.

6 Conclusion

6.1 Summary of findings

This thesis has examined the information dynamics of the 2023 banking crisis with an event study of 48 publicly traded U.S. banks. Four findings come out of it.

First, all five crisis windows contain large negative abnormal returns on naive tests. The window around the SVB closure, which also spans the disclosure and the systemic-risk announcement, produces a cumulated CAAR of -22.31% and a mean bank-level CAR of -21.78% (Section 5.2 reconciles the two), with extreme losses at the failed banks (SIVB -89.0%, FRC -104.5%, SBNY -33.8%).

Second, the cross-sectional variation in CARs over that window is significantly explained by the uninsured deposit ratio ($\beta = -0.0115$, $p = 0.015$ in the preferred full specification, and -0.0112, $p = 0.004$ in the parsimonious one). Equity returns therefore moved with exposure to run-prone funding. Bank size is insignificant, and the three largest banks show a flight-to-safety premium that cannot be separated from their size.

Third, that association survives excluding the three failed institutions ($p = 0.023$), but it is markedly stronger in the lower part of the return distribution and vanishes once the worst-affected fifth is removed, which is the study's central result.

Fourth, Google Trends search interest reached its maximum on March 13, right around the SVB events. However, a panel test of whether the uninsured-deposit penalty was larger on high-attention days gives the predicted sign without significance (interaction -0.0044, bootstrap $p = 0.31$). The co-timing is therefore reported as a description and not as an identified channel (Section 5.4).

Taken together these give one central conclusion. The estimated association between uninsured funding and crisis-period returns sits in the lower tail rather than running evenly across the sample. Whether that comes from a genuine fragility threshold or from the influence of a few extreme outcomes is not settled here, because the formal tests are inconclusive and the association disappears once the worst-affected fifth is removed. Two qualifications shape the finer reading. The aggregate sector-wide reaction cannot be distinguished from zero under the dependence-adjusted test on which the conclusion rests, although a complementary portfolio specification does reject, at roughly three-fifths of the magnitude the naive average suggests. And the tail pattern fits global-games models in which reactions to fragility are sharply non-linear near a threshold, but it fits the influence of a handful of extreme outcomes just as well, so the threshold reading stays suggestive rather than established.

The central claim is meant to be precise, and four boundaries fix it. The estimates are market-pricing relationships and not causal effects, because the design has no exogenous variation in deposit-insurance coverage and no untreated control group, and because the uninsured deposit ratio may stand in for unobserved dimensions of bank quality. The association sits in the lower part of the return distribution rather than running evenly, so there is no coefficient on offer for the typical bank, and that concentration is the substance of the contribution rather than a caveat to it. The information channel is documented but not identified, because the search series is worldwide and aggregate and the interaction of Section 5.4 picks up attention together with the severity of the news. And the aggregate result is indeterminate rather than negative evidence, because the absence of a detectable sector-wide effect does not show that none existed. The panic-versus-fundamentals debate therefore stays open. The evidence here leans towards the intermediate, global-games position, but it does not settle the question.

6.2 Policy implications

First, the uninsured deposit channel suggests that coverage limits deserve another look now that the information environment has changed. The $250,000 limit of the FDIC was designed for retail depositors and leaves corporate and venture-backed balances largely uninsured, and that is what creates the first-mover advantage which makes self-fulfilling runs possible. Goldstein and Pauzner (2005, 1310) identify the offsetting distortion on the other margin: a single bank does not bear the full social cost of a run it may generate, so insurance leads each bank to promise a short-term payment above the social optimum. Coverage limits therefore trade one distortion against another, and the uninsured share is the margin where the two meet: it is the funding that carries no subsidy, so it is where market discipline has to work. One direction follows, although this design estimates market repricing and not welfare. A tiered, risk-based insurance assessment would price the externality directly, with banks funded disproportionately by runnable uninsured deposits paying non-linearly higher premiums rather than relying on ex-post exceptions.

Second, the speed of the 2023 runs, which mobile banking and social media made possible, suggests that a supervisory toolkit built around slower information transmission may no longer be adequate. The Basel III Liquidity Coverage Ratio is calibrated to a 30-day horizon with fairly mild wholesale-outflow assumptions, and it looks mismatched to the speed of digital runs, since SVB lost roughly a quarter of its deposits within hours. There is a case for making the outflow-rate assumptions dynamic, so that they scale with the uninsured-deposit concentration of a bank, and for adding an intraday liquidity stress test at high-uninsured institutions. Whether either would improve on the current calibration is a question this evidence raises without answering.

Third, and much the weakest of the three, the G-SIB coefficient ($\beta = +0.328$, $p = 0.044$, or roughly 33 percentage points of CAR) is consistent with too-big-to-fail expectations remaining a force in bank stock pricing. It rests on three banks, so as Section 5.3 explains it cannot establish that the premium attaches to the designation rather than to scale or diversification, and whether the implicit subsidy distorts competition is a question this design raises but cannot settle.

6.3 Limitations and future research

Three limitations point most directly to future work. Whether the tail concentration comes from a genuine fragility threshold cannot be settled here, because the parametric tests are null and, with only four banks above the cut-off, have little power to tell a threshold from a few severe outcomes. Testing that would need samples with more high-uninsured institutions. The specification also omits variables correlated with uninsured funding, above all unrealised securities losses and hedging, so adding such controls on consolidated holding-company data (Section 4.3) is the most direct extension. And a difference-in-differences design around regulatory thresholds could give the causal identification this design lacks.

7 References

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A Core code and reproducibility

The complete, runnable R project, meaning all scripts, the input data and a README, is provided as supplementary material in a public repository: `https://github.com/diyorahmad71/Information-Dynamics-and-Banking-Run`. The excerpts below reproduce only the core functions that implement the methodology of Chapter 3, with annotation. Routine package-loading, data-download and plotting code is omitted for concision.

A.1 Market-model estimation

The normal-return benchmark is estimated by OLS over the $[-210, -11]$ estimation window, returning α, β and the residual standard deviation $\hat{\sigma}$.

```
market_model <- function(ticker, evt_date) {
  ei <- which.min(abs(returns_df$Date - evt_date))
  est <- returns_df[max(1, ei+EST_WIN[1]):(ei+EST_WIN[2]), ] %>%
           select(Date, R_i = all_of(ticker), R_m = Market) %>% drop_na()
  mod <- lm(R_i ~ R_m, data = est) # OLS over [-210, -11]
  list(alpha = coef(mod)[1], beta = coef(mod)[2],
       sigma = summary(mod)$sigma, n = nrow(est))
}
```

A.2 Abnormal returns and CARs

Abnormal returns are the realised log return minus the market-model prediction over the $[-2, +2]$ window. Day labels are the true trading-day offset from the event (`ew_i - ei`), assigned before missing rows are dropped, so gaps in delisted-bank data cannot mislabel the window.

```
event_study <- function(ticker, evt_date) {
  mm <- market_model(ticker, evt_date)
  ei <- which.min(abs(returns_df$Date - evt_date))
  ew_i <- max(1, ei + EVT_WIN[1]):min(nrow(returns_df), ei + EVT_WIN[2])
  ew <- returns_df[ew_i, ] %>%
            mutate(day = ew_i - ei) %>% # offset, robust to gaps
            select(Date, day, R_i = all_of(ticker), R_m = Market) %>%
            drop_na() %>%
```

```
        mutate(AR = R_i - (mm$alpha + mm$beta * R_m))
  list(CAR = sum(ew$AR), sigma = mm$sigma, n_est = mm$n, n_days = nrow(ew))
}
```

A.3 Aggregation and the full Patell (1976) test

Each bank's CAR is standardised by its prediction-error standard deviation, with the finite-sample degrees-of-freedom correction, before aggregating to the event-level Z-statistic. The line computing `S_i` is equation (6) of Section 3.2, with `n_days` for n_i and `n_est` for M.

```
patell <- results_summary %>%
  mutate(S_i = sigma * sqrt(n_days * (1 + 1/n_est)), # prediction-error SD
         SCAR = CAR / S_i,
         v_i = (n_est - 2) / (n_est - 4)) %>% # d.o.f. adjustment
  group_by(event_id) %>%
  summarise(Patell_Z = sum(SCAR) / sqrt(sum(v_i)))
```

A.4 Cross-sectional regression and clustering-robust inference

The cross-sectional regression uses HC3 robust standard errors. The calendar-time portfolio forms one equal-weighted daily portfolio of all banks in the window and tests its mean abnormal return, which is robust to the cross-sectional dependence from common event dates.

```
# Cross-sectional regression with HC3 robust SEs
m_pars <- lm(CAR ~ uninsured_dep_pct + log(assets_bn), data = reg_df)
coeftest(m_pars, vcov = vcovHC(m_pars, type = "HC3"))

# Calendar-time portfolio (clustering-robust): one equally weighted
# portfolio per calendar day, regressed on the market return. The
# intercept is the average daily abnormal return. H0: alpha_p = 0.
port <- ew_all %>%
  group_by(Date) %>%
  summarise(port_R = mean(R_i), .groups = "drop") %>%
  left_join(select(returns_df, Date, Market), by = "Date")
ct_reg <- lm(port_R ~ Market, data = port)
coeftest(ct_reg, vcov = NeweyWest(ct_reg, lag = 2)) # HAC standard errors

# Attention panel: collapse to ONE row per bank-date before estimating,
```

```
# because the overlapping March windows would otherwise repeat bank-days.
panel_df <- panel_raw %>% distinct(ticker, Date, .keep_all = TRUE)
m <- feols(AR ~ uninsd_x_info | ticker + Date, panel_df, vcov = "twoway")
boottest(m, param = "uninsd_x_info", clustid = "Date", B = 999)
```

The remaining procedures follow the same structure and are provided in full in the repository. They are the multi-source data download with local-CSV fallback, the Fama-French three-factor robustness check, the leave-one-out and bootstrap analyses, the Kolari-Pynnönen dependence adjustment, and all figures.

Two entry points are provided for a reader who wants to check the reported figures rather than read the code. `RUN_EVERYTHING.R` reproduces the whole pipeline end to end from the committed inputs. `12_verify_n48.R` recomputes the quantities reported in Chapters 4 and 5 and compares them against the values printed in this thesis, so any individual figure can be checked without running the full analysis.

B Full-sample descriptive statistics

This appendix reports descriptive statistics for the full 48-bank cross-section used in the Event 2 regressions. The characteristics come from FDIC Call Reports, Q4 2022, and CAR is the market-model cumulative abnormal log return over the $[-2,+2]$ window.

Variable	Mean	SD	Min	Median	Max
CAR at Event 2 (%)	−21.8	22.1	−104.5	−14.5	−1.9
Uninsured deposits (%)	45.4	13.3	32.4	41.7	94.3
Total assets ($B)	233.6	608.9	4.8	45.4	3201.9
Tier 1 leverage (%)	9.0	0.7	7.7	9.1	11.4
Loan-to-deposit (%)	76.3	14.9	42.0	76.2	106.0

Table B.1: Summary statistics ($N = 48$).

	CAR	Uninsured	log(Assets)	Tier 1	LTD
CAR	1.00	−0.67	−0.23	0.32	−0.10
Uninsured	−0.67	1.00	0.36	−0.33	−0.21
log(Assets)	−0.23	0.36	1.00	−0.64	−0.22
Tier 1	0.32	−0.33	−0.64	1.00	0.17
LTD	−0.10	−0.21	−0.22	0.17	1.00

Table B.2: Correlation matrix of regression variables ($N = 48$). CAR is the Event 2 cumulative abnormal log return. The remaining variables are the Q4 2022 characteristics defined in Table 2.

Table B.3: Bank characteristics, all 48 institutions (Q4 2022). Note: the bank-by-bank listing summarised by group in Table 2. Definitions are as in Table 2. Source: FDIC BankFind Q4 2022, with PacWest sourced from its Q4 2022 earnings release (Section 4.3).

Ticker	Group	Uninsured %	Assets $B	Tier 1 %	LTD %
BAC	Large	41.7	2418.5	7.68	50.4
BANR	Regional	37.9	15.8	9.40	74.5
BOKF	Regional	48.2	47.8	8.60	72.1
CATY	Regional	38.1	22.6	9.70	74.3
CBSH	Regional	56.7	31.7	9.20	61.7
CFG	Regional	41.9	226.4	8.85	86.2
CMA	Regional	59.3	85.5	9.01	71.9
COLB	Regional	38.6	52.1	8.90	81.0
CVBF	Regional	41.7	16.0	9.90	42.1
EBC	Regional	32.7	13.4	9.20	63.2
EWBC	Regional	47.2	64.1	9.15	85.8

Ticker	Group	Uninsured %	Assets $B	Tier 1 %	LTD %
FFBC	Regional	32.8	16.9	9.30	80.5
FFIN	Regional	50.7	13.1	9.80	49.3
FHN	Regional	45.1	78.7	9.20	90.5
FITB	Regional	46.5	206.3	8.45	72.3
FRC	Failed	67.3	212.6	8.51	94.1
GABC	Regional	37.5	5.8	9.40	66.8
GBCI	Regional	42.7	26.6	9.60	73.8
HAFC	Regional	37.6	7.3	9.50	96.7
HBAN	Regional	44.2	182.3	8.12	79.5
HWC	Regional	38.5	35.2	9.80	79.2
IBCP	Regional	34.7	4.8	9.20	72.3
INDB	Regional	37.8	19.3	9.20	86.8
JPM	Large	59.5	3201.9	8.30	46.1
KEY	Regional	48.2	187.6	8.78	81.8
ONB	Regional	35.7	46.5	9.10	87.9
OZK	Regional	32.4	27.7	11.40	96.6
PACW	Regional	71.1	44.3	9.70	84.4
PB	Regional	37.6	37.9	9.00	56.8
RF	Regional	34.8	154.2	8.80	71.7
RNST	Regional	34.5	17.0	9.10	85.6
SBNY	Failed	89.3	110.4	8.79	83.9
SFNC	Regional	34.8	26.5	9.30	66.1
SIVB	Failed	94.3	209.0	7.96	42.0
TCBI	Regional	54.9	36.7	9.50	79.4
TFC	Large	41.8	546.0	8.54	76.0
TRMK	Regional	37.1	16.8	9.20	73.6
UBSI	Regional	34.4	29.4	9.10	91.4
UMBF	Regional	54.0	37.7	8.10	68.9
USB	Large	52.2	585.1	8.05	71.2
VLY	Regional	39.8	57.5	8.40	98.2
WAFD	Regional	35.7	21.6	9.30	106.0
WAL	Regional	52.6	67.7	8.22	98.7
WBS	Regional	42.7	71.2	8.50	91.5
WFC	Large	39.3	1717.5	8.34	63.8
WSFS	Regional	34.6	16.1	9.30	74.1
WTFC	Regional	40.1	52.9	8.95	86.3
ZION	Regional	49.2	89.5	7.65	76.5

C Supplementary tables

Ticker	Alpha ($\hat{\alpha}$)	Beta ($\hat{\beta}$)	R^2	Est. Obs.	Resid. Std Dev.
SIVB	−0.0024	1.74	0.37	200	0.0339
SBNY	−0.0033	1.77	0.57	200	0.0227
FRC	−0.0006	1.32	0.57	200	0.0171
ZION	−0.0003	1.11	0.59	200	0.0138
WAL	0.0000	1.45	0.64	200	0.0162
PACW	−0.0006	1.40	0.62	200	0.0163
CMA	−0.0005	1.03	0.50	200	0.0152
KEY	−0.0001	1.13	0.59	200	0.0139
RF	0.0007	0.97	0.53	200	0.0136
JPM	0.0008	0.85	0.55	200	0.0114
BAC	−0.0002	0.94	0.56	200	0.0123
WFC	0.0004	0.93	0.52	200	0.0131
USB	0.0001	0.78	0.45	200	0.0128
TFC	0.0001	0.99	0.57	200	0.0128

Table C.1: Market Model Estimates, Estimation Window $[-210,-11]$ (representative subset of 14 banks). Note: this table reports a representative subset spanning the size and volatility range of the sample, not all 48 banks. The full set is provided in the supplementary repository. Alpha and Beta estimated by OLS over the estimation window. R^2 is the coefficient of determination. Est. Obs. = number of valid daily observations in the estimation window. Resid. Std Dev. = standard deviation of OLS residuals, used as the denominator in the CAR t-statistic. All fourteen estimations use the full 200 trading days of the window, as does every other bank in the sample: returns are computed on each series' own observed prices, so a missing day for one bank does not remove an observation from any other. PACW and CMA are estimated here on the secondary-provider series described in Section 4.1, the same series from which their Event 2 CARs in Table C.4 are computed. Across all 48 banks the market-model beta runs from 0.27 (FHN) to 1.77 (SBNY), the R^2 from 0.25 to 0.64, and the residual standard deviation from 0.0069 (FHN) to 0.0339 (SIVB). The fourteen shown span most of that range.

Variable	VIF	Assessment
Uninsured Deposits (%)	1.38	No multicollinearity
log(Assets)	6.22	Moderate, below the conventional cutoff
Tier 1 Leverage Ratio	1.77	No multicollinearity
Loan-to-Deposit Ratio	1.36	No multicollinearity
Regulatory Exempt Dummy	3.31	No multicollinearity
G-SIB Dummy	2.65	No multicollinearity

Table C.2: Variance Inflation Factors (VIF), full cross-sectional model. Note: VIF > 10 indicates severe multicollinearity, and VIF > 5 warrants caution. Computed via auxiliary OLS regressions of each predictor on all remaining predictors, on the six-regressor specification and using log(Assets) as it enters the estimated model. Entering total assets in levels instead would raise the size variable to 13.2 and the G-SIB dummy to 11.4, since in levels the dummy is close to a deterministic function of the three largest values. The logarithmic transformation used throughout removes that near-collinearity, which is one reason it is preferred. The imprecision of the G-SIB coefficient in Section 5.3 therefore reflects the three-bank size of the treated group rather than collinearity with size.

Test	Statistic	p-value	Verdict
Breusch-Pagan (heterosk.)	LM $= 21.16$, df $= 6$	0.002	Heteroskedasticity confirmed, HC3 robust SEs appropriate
Jarque-Bera (normality)	JB $= 14.20$, df $= 2$	< 0.001	Non-normal residuals (skew $= -0.626$, excess kurt. $= 2.352$)
Cook's Distance (influence)	Max $D = 0.824$ (SBNY)	—	6 obs exceed $4/n$ threshold (SBNY, FRC, SIVB, WAL, JPM, OZK). None exceed the stricter $D > 1$ cutoff

Table C.3: Regression diagnostic tests. Note: the Breusch-Pagan statistic and the variance inflation factors of Table C.2 are computed on the six-regressor specification that adds the G-SIB and regulatory-exemption indicators to the four variables of Table 5, which is why the Breusch-Pagan statistic carries six degrees of freedom and Table C.2 reports six variance inflation factors. This is the widest specification estimated, so it is the conservative choice for detecting heteroskedasticity and collinearity. The Cook's distances, by contrast, are computed on the four-regressor full specification of Table 5, since influence on the coefficients actually reported is the quantity of interest, and so is the Jarque-Bera statistic. Breusch-Pagan LM statistic distributed $\chi^2(6)$ under H_0 of homoskedasticity, and $\chi^2(6) = 21.16$ corresponds to $p = 0.002$. Jarque-Bera statistic distributed $\chi^2(2)$ under H_0 of normality, with the third and fourth moments standardised by the sample standard deviation. Standardising by the population second moment instead returns JB $= 16.6$ (skew -0.643, excess kurtosis 2.581) on the same residuals, and on the six-regressor residuals used for the Breusch-Pagan test the same two conventions give JB $= 20.4$ and 23.6 respectively. Normality is rejected on all four. Cook's distance threshold $= 4/n = 4/48 = 0.083$. For comparison, the stricter absolute cutoff of $D > 1$ sometimes recommended for flagging genuinely dominant observations (Cook and Weisberg, 1982) is breached by no bank in the full sample (maximum $D = 0.824$ for SBNY).

Table C.4: Cumulative Abnormal Returns at Event 2, all 48 banks (sorted from most to least negative). Note: all CARs are cumulative log-return abnormal returns over the $[-2,+2]$ window. Log returns are additive across days, so a cumulative value below -100% is admissible and does not violate limited liability. The final column gives the equivalent simple cumulative return, $e^{\text{CAR}}-1$, computed from the log CAR beside it so the conversion checks row by row: First Republic's -104.5% is -64.8% in simple terms. The banks named in the text of Section 5.1 are all listed here. PacWest and Comerica are included, and their series were reconstructed from secondary providers as described in Section 4.1. The treatment of the halted banks is as in Section 3.2. Source: author calculations.

Ticker	Group	CAR, log-return units (%)	Equivalent simple return (%)
FRC	Failed	−104.5	−64.8
SIVB	Failed	−89.0	−58.9
WAL	Regional	−85.3	−57.4
PACW	Regional	−69.1	−49.9
CMA	Regional	−38.9	−32.2
ZION	Regional	−37.4	−31.2
KEY	Regional	−34.7	−29.3
SBNY	Failed	−33.8	−28.7
TFC	Large	−30.8	−26.5
UMBF	Regional	−27.5	−24.0
FHN	Regional	−26.6	−23.4
HBAN	Regional	−25.7	−22.7
FITB	Regional	−24.4	−21.7
COLB	Regional	−22.5	−20.1
WBS	Regional	−22.0	−19.7
EWBC	Regional	−21.9	−19.7
EBC	Regional	−20.7	−18.7
USB	Large	−18.3	−16.7
CFG	Regional	−17.8	−16.3
HWC	Regional	−17.5	−16.1
CVBF	Regional	−17.5	−16.1
WSFS	Regional	−15.3	−14.2
WTFC	Regional	−14.8	−13.8
IBCP	Regional	−14.6	−13.6
RF	Regional	−14.4	−13.4
TCBI	Regional	−14.0	−13.1
HAFC	Regional	−12.6	−11.8
BAC	Large	−12.0	−11.3
BANR	Regional	−11.5	−10.9
BOKF	Regional	−11.0	−10.4

Ticker	Group	CAR, log-return units (%)	Equivalent simple return (%)
PB	Regional	−10.8	−10.2
OZK	Regional	−10.5	−10.0
CATY	Regional	−10.5	−10.0
VLY	Regional	−10.0	−9.5
INDB	Regional	−9.8	−9.3
ONB	Regional	−9.6	−9.2
GABC	Regional	−9.6	−9.2
WFC	Large	−8.8	−8.4
FFIN	Regional	−8.6	−8.2
TRMK	Regional	−8.3	−8.0
SFNC	Regional	−7.8	−7.5
GBCI	Regional	−6.3	−6.1
RNST	Regional	−5.9	−5.7
WAFD	Regional	−5.8	−5.6
FFBC	Regional	−5.4	−5.3
CBSH	Regional	−5.1	−5.0
UBSI	Regional	−4.5	−4.4
JPM	Large	−1.9	−1.9

Ticker	Market-model CAR %	Excess-CAPM CAR %	FF3 CAR %	Diff (pp)
FRC	−104.5	−103.8	−102.1	+1.7
SIVB	−89.0	−88.6	−88.4	+0.2
WAL	−85.3	−84.6	−79.6	+5.0
PACW	−69.1	−68.3	−61.7	+6.6
CMA	−38.9	−38.4	−33.3	+5.1
ZION	−37.4	−36.8	−31.4	+5.4
KEY	−34.7	−34.1	−29.4	+4.7
SBNY	−33.8	−33.1	−30.3	+2.8
TFC	−30.8	−30.3	−26.1	+4.2
UMBF	−27.5	−27.1	−24.7	+2.4
FHN	−26.6	−26.5	−25.2	+1.3
HBAN	−25.7	−25.3	−20.8	+4.5

Table C.5: Market model, excess-return CAPM and Fama-French three-factor CARs at Event 2, for the twelve most severely affected banks. Note: the first column reproduces the raw-return market-model CAR of Table C.4. For comparability with the FF3 model, the second column re-estimates the single-factor model on excess returns ($R_i - R_f$ regressed on $R_m - R_f$) over the same estimation window, which is why it differs marginally from the first. Diff (pp) = FF3 CAR − excess-CAPM CAR, in percentage points. Within these twelve banks the FF3 ordering differs from the market-model ordering at two adjacent pairs only, KEY and SBNY and UMBF and FHN. Across the full 48 the two series correlate at 0.998 and 29 banks change rank, almost all by a single place. The full 48-bank comparison is provided in the supplementary repository.

Sample	β (Uninsured Dep.)	p	N
All banks	-0.0112	0.004	48
Excluding the three failed banks	-0.0103	0.023	45
Excluding the three most negative CARs	-0.0064	0.064	45
Excluding the four most negative CARs	-0.0039	0.004	44
Excluding the bottom decile	-0.0032	0.028	43
Excluding the bottom quintile	-0.0002	0.911	38

Table C.6: Progressive trimming from the lower tail. Note: each row re-estimates the parsimonious specification of Table 5, that is CAR on the uninsured deposit ratio and log(assets), on the sample described, with HC3 standard errors. The three failed banks are SIVB, SBNY and FRC. The three most negative CARs are FRC, SIVB and WAL, and the fourth adds PACW, so the second and third rows drop the same number of banks but not the same banks. This four-bank group is not the one used in Section 5.2, which pairs the three failures with Western Alliance rather than with PacWest. The step from the third row to the fourth lowers both the coefficient and its p-value, which Section 5.5 explains. Source: author's calculations.

C.1 Supplementary robustness detail (from Section 5.5)

A Fama-French three-factor (FF3) robustness check confirms the CARs are not an artefact of the single-factor benchmark. Across all 48 banks the market-model and FF3 CARs at Event 2 correlate at 0.998 and preserve the broad ordering of banks. FF3 CARs are on average 5.0 percentage points less negative (mean -16.82% against the market-model mean of -21.78%), consistent with regional banks' positive size and value loadings, while the two largest CARs move very little, Silicon Valley Bank by 0.2 and First Republic by 1.7 percentage points against the excess-return CAPM baseline of Table C.5, and by 0.6 and 2.4 against the market model. The ordering is preserved in the tail but not exactly throughout. 29 of the 48 banks change rank, almost all by a single place, and as Table C.5 shows the reshuffling within the twelve worst-affected banks is confined to two adjacent pairs. What matters more is that re-estimating the parsimonious cross-sectional regression on the FF3 CARs leaves the result intact: the uninsured-deposit coefficient is -0.0116 ($p = 0.003$), statistically and economically indistinguishable from the market-model estimate of -0.0112. The main finding is therefore robust to the choice of normal-return benchmark. Daily Fama-French factors are taken from the Kenneth French Data Library and the three-factor model is estimated over the same $[-210, -11]$ window as the market model. That FF3 CARs are systematically less negative is itself consistent with regional banks loading positively on the SMB and HML factors, so that the single-factor model attributes to abnormal performance some return variation the three-factor model assigns to systematic size and value exposure. Because the two series correlate at 0.998, the substantive conclusions are unchanged. Table C.5 reports the comparison for the twelve most affected banks, and the full 48-bank comparison is provided in the supplementary repository.

C.2 Threshold tests and the unequal event windows (from Section 5.5)

To test the threshold interpretation directly, the parsimonious model was augmented with a quadratic term (uninsured^2) and then with an interaction between the uninsured ratio and an indicator for uninsured deposits above 60%, a cut-off that isolates the three failed banks together with PacWest. Neither the quadratic term ($p = 0.87$) nor the threshold interaction ($p = 0.77$) is statistically significant. With only four banks above the cut-off, however, these parametric tests have very little power to distinguish a genuine discontinuity from four joint outliers. Their failure to reject should not be read as evidence against non-linearity.

Abnormal returns are cumulated only over days on which a bank actually traded, so two institutions contribute shorter windows than the rest (Section 3.2): SIVB contributes two trading days and SBNY three. Two checks address this. Re-estimating the parsimonious specification on the complete-window banks alone ($N = 46$) gives -0.0142 ($t = -2.90$, $p = 0.006$). Replacing the dependent variable with the mean daily abnormal return, which is invariant to window length, gives -0.0041 per day ($t = -2.06$, $p = 0.045$), or -0.0206 over the five-day window. Both keep their sign and both are significant at the 5% level. On a common five-day basis both are larger in absolute value than the full-sample -0.0112, by 27 and 84 per cent respectively. Nothing in the main result therefore depends on the unequal windows.

C.3 Diagnostic detail (from Section 5.3)

This appendix expands the four diagnostics summarised in Section 5.3 and reported in Tables C.2 and C.3, giving the underlying statistics and the reasoning behind each. Nothing here revises the summary. It supplies the detail behind it.

The VIF analysis shows moderate collinearity between log(Assets) (VIF = 6.22) and the other size-related regressors, with the G-SIB dummy at 2.65. Neither exceeds the conventional cutoff of 10, so the coefficients on these two variables are separately estimable, though the G-SIB coefficient is imprecise because the indicator is switched on for only three banks. Estimability is not identification. As Section 5.3 sets out, those three banks are also the three largest in the sample, so the premium cannot be attributed to the designation rather than to sheer size, and the abstract's description of it as confounded with size is the one that governs. Had total assets entered in levels rather than logs, the corresponding figures would be 13.2 and

11.4, because in levels the G-SIB dummy is close to a deterministic function of the three largest values. This is one reason the logarithmic transformation is used throughout. The uninsured deposit coefficient, the primary variable of interest, is unaffected on either measure (VIF $= 1.38$ in logs, 1.33 in levels).

The rejection of normality reported in Table C.3 is driven by a negative skew of -0.626 and excess kurtosis of 2.35, consistent with a few banks holding extreme negative CARs (SIVB, FRC). With $N = 48$ asymptotic normality cannot be assumed, which is why inference rests on the HC3 sandwich estimator, which does not require normality though it remains approximate at this sample size, and on the permutation sensitivity test of Section 3.4, which relaxes the independence assumption though not the exchangeability one it puts in its place.

The individual Cook's distances behind Table C.3 are SBNY (0.824), FRC (0.361), SIVB (0.306), WAL (0.230), JPM (0.203) and OZK (0.109), computed on the four-regressor specification of Table 5. PACW and CVBF both fall just below the $4/n = 0.083$ screen, at 0.078 and 0.078 to three decimals, though they are not identical at higher precision. Two features of this ranking matter for the reading of the cross-section. SBNY exceeds the absolute $D > 0.5$ guideline as well as the $4/n$ screen, so it is influential on any criterion, while PacWest is not flagged at all in the full sample. That second point should not be misread. Influence is specific to the sample in which it is measured, and in the reduced subsample of Section 5.5 from which PacWest is dropped its Cook's distance rises to 1.04, making it by a clear margin the most influential observation there. The influence is concentrated in the institutions that failed or were severely repriced, which is why the failed-bank exclusion, the bank-level bootstrap and the Huber estimator of Section 5.5 are the appropriate responses, and why the cross-sectional result is read as concentrated among those institutions rather than as a property of the sample as a whole.

Statement of Authorship

I hereby affirm that I have written the above thesis independently and have used no sources or aids other than those indicated; that the submitted work has not previously been submitted for examination at any other university; and that it has not been published, either in whole or in part. I have clearly indicated all direct quotations and passages paraphrased from other works as such.

Violations of this declaration may be considered an attempt at deception.

Date:

Place:

Signature: